



Agentic Architecture

Agentic AI Architecture COE Structure, Operating Model

February 2026



Table of Contents

- 1. Executive Summary**
- 2. Operating Model Structure**
- 3. Organization**
- 4. Platform Journey and Platform MVP**
- 5. Appendix**
 - A1. Example Factory Delivery Model**
 - A2. Role Details**



Executive Summary



Executive Summary Narrative

Agentic AI adoption is accelerating across retail, with peers moving to platform-based models. However, industry data shows a high failure rate due to fragmented execution, unclear ownership, and lack of cost discipline.

To capture value at scale and avoid these pitfalls, we recommend establishing a centralized Enterprise AI Hub to drive standards, reuse, governance, and ROI realization across Agentic AI solutions.

Operating Model

- Establish an Enterprise AI Hub to build and govern core Agentic AI platform capabilities
- Centralize solution engineering while partnering with Value Streams to deploy use cases
- Embed FinOps and value management to ensure ROI transparency and cost discipline
- Align closely with enabling functions (Data, Cloud, Infrastructure, Security, Legal)

Organization & Delivery Model

- Launch with a Platform MVP aligned to priority use cases (CCAS, GEO, Conversational Search)
- Design and hire for the steady-state operating model from the outset
- Recruit key strategic and technical leadership roles to anchor the Hub
- Leverage offshore delivery for standardized, repeatable functions (Ops, platform support, enabling IT)

How we Get Started

Platform MVP

- Decide on the three MVP scope options based on speed-to-value ambition
- Clear decision point on investment and pace
- Co-build initial MVP with partner while parallel pathing hiring and upskilling

Targeted Talent Acquisition along with Upskilling Internal Resources

- ~20 new AI roles identified with 8 external hires and 12 upskilled internal resources
- ~40 critical AI skills identified (functional + technical)

Expected Outcomes

- Scalable and secure enterprise Agentic AI foundation
- Standardized, reusable and interoperable Agentic AI capabilities
- Faster time to value for implementing use case solutions
- Transparent value tracking and cost controls
- Long-term AI talent and capability development within Costco

Decisions Required

1. Confirm centralized AI Hub operating model & organization
2. Select Platform MVP scope and the speed with which you want to move



Operating Model Structure

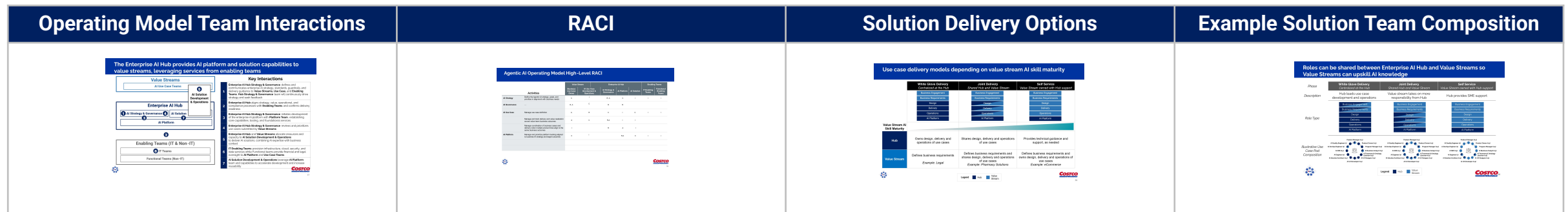
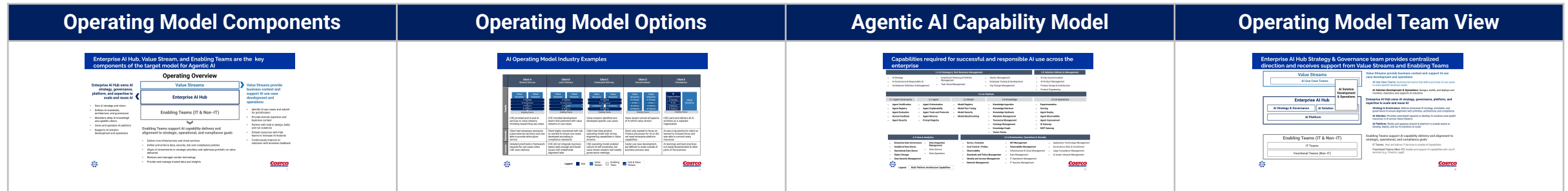


Operating Model Structure Section Overview

This section outlines the operating model required to design, deliver, and scale Agentic AI capabilities across the enterprise.

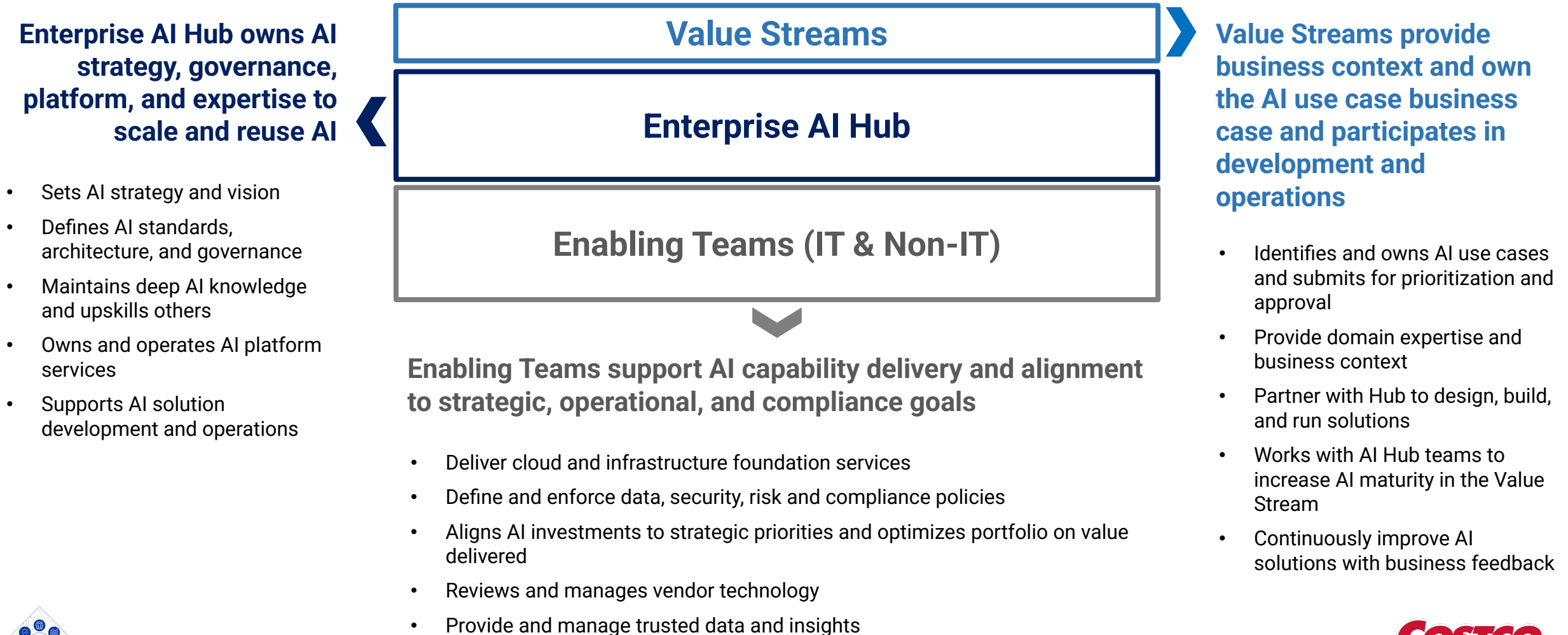
Recommendations Summary

1. Establish a centralized Agentic AI Hub that has full ownership of the platform
2. Initially Agentic AI Hub builds a centralized solution capability and has shared solution development responsibilities with Value Streams to deliver on use cases
3. Over time evolve to a federated model where value streams own solution development



Enterprise AI Hub, Value Stream, and Enabling Teams are the key components of the target model for Agentic AI

Operating Overview



AI Operating Model Archetypes



Spectrum of AI operating models to best balance AI strategy, capability maturity, and organizational autonomy

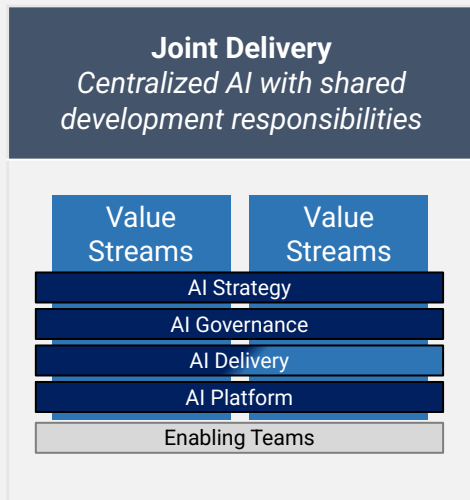
Consideration for Costco

Example	Shared Service <i>Centralized AI to serve enterprise</i>	Joint Delivery <i>Centralized AI with shared development responsibilities</i>	Federated Delivery <i>Centralized AI with value streams owning development</i>	Decentralized <i>Value stream ownership and execution of AI</i>	Standalone <i>Separate organization to deliver AI end-to-end</i>
Details	AI Hub owns AI strategy, governance, delivery, and platform	- AI Hub owns AI strategy, governance, and platform - COE and Value Streams share AI delivery	- AI Hub owns strategy, governance, and platform - Value Streams own AI delivery	Value Streams own AI strategy, governance, delivery, and platform	AI Hub is a separate organization that owns AI strategy, governance, delivery, and platform
Pros	- Concentrated talent and skills - Greater ability to identify and implement re-usability	- Shared accountability for delivery - Increased Value Stream context	- Reduces delivery bottleneck - Value Streams accountable for scaling development	- Faster speed for Value Stream-specific use cases - Fewer dependencies on central teams	- Accelerated speed because of focus - Freedom from enterprise constraints
Cons	- Demand bottleneck, slower speed - Greater effort to gain Value Stream context	- Coordination required within development team - Potential ambiguity for delivery accountability	- Requires strong governance effort - Value Streams have varying development capabilities	- Duplicated effort leading to higher overall spend - Inconsistent standards across value streams	- Potential disconnect from business needs / strategy - More coordination effort required with Value Streams



Recommendation is to start with a Joint Delivery model and as Value Streams and partners have capabilities - move to a Federated Model

◆ Recommended Initial State



- Sets strong AI foundation and enables control over technical architecture and standards
- Increases use case development speed by partnering with value stream teams
- Develops enterprise platform concurrently and facilitates reusability

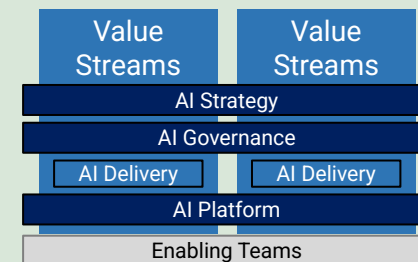
To move from recommended initial to target state, the following must be true:

- ✓ Strategy defined and aligned with stakeholders
- ✓ Knowledge transferred from AI Hub to value stream teams
- ✓ Compliance and processes established with enabling teams
- ✓ AI platform developed and operational
- ✓ Self-service capabilities including mature knowledge management
- ✓ Preferred partners are established that are familiar with the AI Platform

Legend ■ Hub ■ Value Stream ■ Enabling Team

★ Desired Target State

Federated Delivery
Centralized AI with value streams owning development



- AI Hub controls and monitors strategy, governance and adherence to standards
- Value Streams that do not have delivery capabilities can leverage preferred partners
- AI Hub will provide professional services to review designs, certify solutions, and get involved as needed
- Supports Costco's culture of autonomy and reduces desire for shadow IT



Capabilities required to enable Agentic AI successfully and responsibly across the enterprise

Agentic AI Capability Model

1.0 AI Strategy & Tech Business Management

Establishes AI technology strategy and standards and governs investments and organizational behavior to ensure alignment with organizational priorities and responsible behaviors.

2.0 Solution Delivery & Management

Designs, delivers, and manages AI use cases end-to-end, ensuring solutions are built, deployed, and continuously improved to deliver measurable business value.

3.0 AI Platform

Creates shared enterprise platform that supplies the core orchestration, tooling, integration, technical governance, and operational services needed to enable AI.

4.0 Data & Insights

Delivers and governs high-quality, trusted data to power AI agents, while providing data and analytics capabilities that inform and continuously shape the enterprise AI strategy.

5.0 Infrastructure, Operations & Security

Provides resilient, secure, and optimized cloud and infrastructure services to continuously run AI solutions responsibly across the enterprise.



Capabilities required for successful and responsible AI use across the enterprise

Legend: **Enabling Capabilities**

1.0 AI Strategy & Tech Business Management

- AI Strategy
- AI Governance & Responsible AI
- Architecture Definition & Management
- Investment Planning & Portfolio Management
- Tech Value Management
- Vendor Management
- Employee Training & Development
- Org Change Management

2.0 Solution Delivery & Management

- AI Use Case Innovation
- AI Product Management
- Product Design & Architecture
- Product Engineering

3.0 AI Platform

3.1 Agent Governance

- Agent Certification
- Agent Registry
- Agent Evaluation
- Human Feedback
- Agent Security

3.2 Agent

- Agent Orchestration
- Agent Explainability
- Agent Tools and Protocols
- Agent Memory
- Prompt Registry

3.3 Model

- Model Registry
- Model Fine Tuning
- Model Security
- Model Benchmarking

3.4 Knowledge

- Knowledge Ingestion
- Knowledge Retrieval
- Knowledge Synthesis
- Metadata Management
- Taxonomy Management
- Ontology Management
- Knowledge Graph
- Vector Stores

3.5 AI Operations

- Experimentation
- Serving
- Agent Deploy
- Agent
- Agent Improvement
- AI Gateway
- MCP Gateway

4.0 Data & Insights

- Enterprise Data Governance
- Analytical Data Stores
- Operational Data Stores
- Object Storage
- Data Security Management
- Data Integration Management
- Data Science
- Data Operations

5.0 Infrastructure, Operations & Security

- Server, Container
- Cost Control / FinOps
- Standards and Policy Management
- Identity and Access Management
- Network Management
- API Management
- Vulnerability Management
- Infrastructure & Cloud Management
- Data Management
- IT Operations Management
- IT Security Management
- Application Technology Management
- Governance, Risk, & Compliance
- Legal Compliance Management
- AI Asset Lifecycle Management



Agentic AI Platform Technical Capabilities are a subset of the AI Capability Model

Level 3 Capability

EXISTING CAPABILITY

CLOUD & INFRA

PLATFORM

DATA

Agent Governance	Agent Certification (Certified 1P and 3p Agents)	Agent Registry (Agent registration – 1P and 3P agents)	Agent Evaluation (LLM as judge, statistical metrics)	
	Human Feedback (Thumbs up/down, structured feedback)	Agent Security (Identity management, Data and Tool controls)		
Agent	Agent Orchestration (no code, low code, pro-code)	Agent Explainability (Telemetry)	Agent Tools and Protocols (MCP, A2A, connectors)	
	Agent Memory (Memory management)	Prompt Registry (Prompt Management)		
Model	Model Registry (Registry of approved Foundation models)	Model Fine Tuning (Domain Adaptation of foundation models)	Model Security (Guardrails, Content Filter)	Model Benchmarking (Identifying the right-fit model for the use case)
Knowledge	Knowledge Ingestion (Semi-structured, unstructured & structured data)	Knowledge Retrieval (Retrieval strategies)	Knowledge Synthesis (Synthetic data generation)	Metadata Management (metadata for unstructured content)
	Taxonomy Management (Create, curate and manage taxonomy)	Ontology Management (Create, curate and manage ontologies)	Knowledge Graph (KG creation using taxonomy and ontology)	Vector Stores (Embedding Persistence)
Data	E Enterprise Data Governance (Data Catalog)	E Analytical Data Stores (Analytical Data Product)	E Operational Data Stores (Near Real time Data Product)	E Object Storage
	E Data Security Management (Masking, Encryption)	E Data Integration Management (ETL, Pub/Sub, CDC)		
Cloud & Infrastructure	E Server, Container (Agent Run Time)	E Cost Control (Tagging, Budgets and Alerts)	E	E Standards and Policy Management (NIST Controls)
	E Identity and Access Management (Identity Access Management)	E Network Management (VPC, Subnets, Routes)	E (Telemetry) API Management (API access)	E Vulnerability Management

AI Operations

E Experimentation
(ML model experiment, Build, deploy, monitor)

E Serving
(ML Inferencing)

Agent Deploy
(Agentic Application development and operations)

Agent
(Model, Application, Agent, Prompts, Ingestion pipelines)

Agent Improvement
(Feedback Integration)

AI Gateway
(Model, Agent)

MCP Gateway
(Tools)

Category	Capability	Level 1	Level 2	Level 3
Agent Governance	Agent Certification			
Agent Governance	Human Feedback			
Agent Governance	Agent Registry			
Agent Governance	Agent Security			
Agent Governance	Agent Evaluation			
Agent	Agent Orchestration			
Agent	Agent Memory			
Agent	Agent Explainability			
Agent	Prompt Registry			
Model	Model Registry			
Model	Model Fine Tuning			
Model	Model Security			
Model	Model Benchmarking			
Knowledge	Knowledge Ingestion			
Knowledge	Knowledge Retrieval			
Knowledge	Knowledge Synthesis			
Knowledge	Metadata Management			
Knowledge	Taxonomy Management			
Knowledge	Ontology Management			
Knowledge	Knowledge Graph			
Knowledge	Vector Stores			
Data	Enterprise Data Governance			
Data	Analytical Data Stores			
Data	Operational Data Stores			
Data	Object Storage			
Data	Data Security Management			
Data	Data Integration Management			
Cloud & Infrastructure	Server, Container			
Cloud & Infrastructure	Cost Control			
Cloud & Infrastructure	Identity and Access Management			
Cloud & Infrastructure	Network Management			
Cloud & Infrastructure	(Telemetry)			
Cloud & Infrastructure	API Management			
Cloud & Infrastructure	Standards and Policy Management			
Cloud & Infrastructure	Vulnerability Management			

Agentic AI Capability Model



Agentic AI Capability Ownership

	 AI Hub	 Value Stream	 Enabling Team
Ownership Definition	Centralized team that provides leadership, best practices, support, and training for AI	Value stream teams that require an AI solution	Functional teams (outside of AI Hub and value streams) that deliver enterprise services and enabling capabilities
Capability Ownership Criteria	• Requires enterprise-wide consistency • High risk or regulatory impact that can apply globally • Benefits from economies of scale • Defines standards and guardrails • Scarce or specialized expertise	• Value stream-specific context required • Focus on use case delivery and iteration • High business value and requires close business collaboration • Consumes standards and provides feedback and standards requests	• Mature and standardized capability that spans enterprise • Requires operational scale • Commodity execution • Can be vendor-managed • Delivered through service requests and measured against SLAs
Examples	• AI Architecture and Standards • Responsible AI	• Agent / AI Use Cases • Product Management	• Cloud Infrastructure Operations • Network and Core Infrastructure



Capabilities required for successful and responsible AI use across the enterprise

Legend  Hub  Value Stream  Enabling Team **Bold:** Enabling Capabilities

1.0 AI Strategy & Tech Business Management

 AI Strategy	 Investment Planning & Portfolio Management	 Vendor Management
 AI Governance & Responsible AI		 Employee Training & Development
 Architecture Definition & Management	 Tech Value Management	 Org Change Management

2.0 Solution Delivery & Management

 AI Use Case Innovation
 AI Product Management
 Product Design & Architecture
 Product Engineering

3.0 AI Platform

3.1 Agent Governance

 Agent Certification
 Agent Registry
 Agent Evaluation
 Human Feedback
 Agent Security

3.2 Agent

 Agent Orchestration
 Agent Explainability
 Agent Tools and Protocols
 Agent Memory
 Prompt Registry

3.3 Model

 Model Registry
 Model Fine Tuning
 Model Security
 Model Benchmarking

3.4 Knowledge

 Knowledge Ingestion
 Knowledge Retrieval
 Knowledge Synthesis
 Metadata Management
 Taxonomy Management
 Ontology Management
 Knowledge Graph
 Vector Stores

3.5 AI Operations

 Experimentation
 Serving
 Agent Deploy
 Agent
 Agent Improvement
 AI Gateway
 MCP Gateway

4.0 Data & Analytics

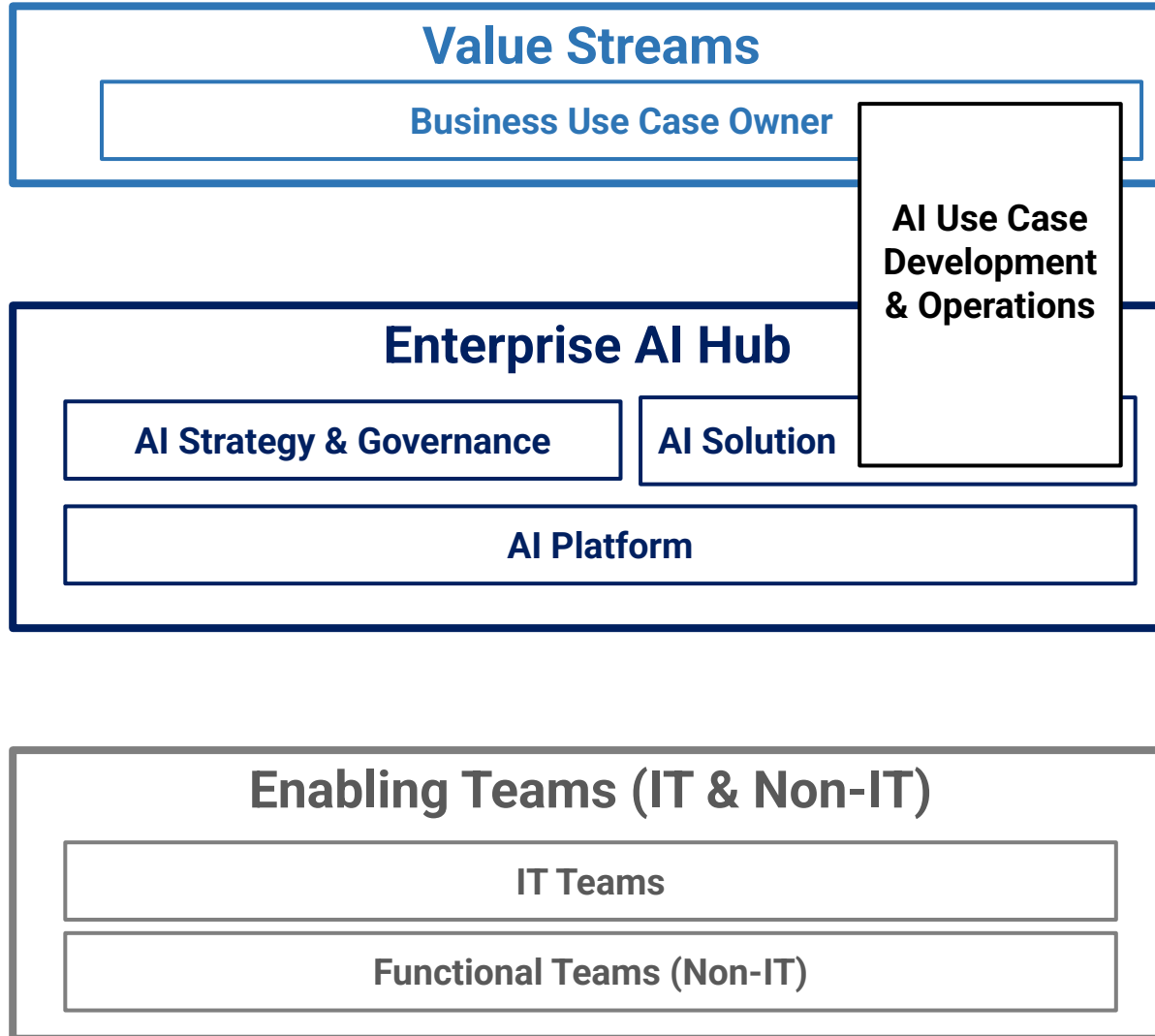
 Enterprise Data Governance	 Data Integration Management
 Analytical Data Stores	
 Operational Data Stores	 Data Science
 Object Storage	 Data Operations
 Data Security Management	

5.0 Infrastructure, Operations & Security

 Server, Container	 API Management	 Application Technology Mgmt.
 Cost Control / FinOps	 Vulnerability Management	 Governance, Risk, & Compliance
	 Infrastructure & Cloud Management	 Legal Compliance Management
 Standards and Policy Management	 Data Management	 AI Asset Lifecycle Management
 Identity and Access Management	 IT Operations Management	
 Network Management	 IT Security Management	



Enterprise AI Hub Strategy & Governance team provides centralized direction and receives support from Value Streams and Enabling Teams



Value Streams provide business context and own the AI use case business case and participates in development and operations

Business Use Case Owner: Business-led teams that define and steer AI use cases to solve specific business needs

AI Use Case Development & Operations: Designs, builds, and deploys and monitors, maintains, and supports AI solutions.

Enterprise AI Hub owns AI strategy, governance, platform, and expertise to scale and reuse AI

Strategy & Governance: Defines enterprise AI strategy, standards, and governance to ensure alignment with priorities, architecture, and compliance

AI Solution: Provides centralized capacity to develop AI solutions and upskill resources in AI across Value Streams

AI Platform: Builds and operates shared AI platform to enable teams to develop, deploy, and run AI solutions at scale

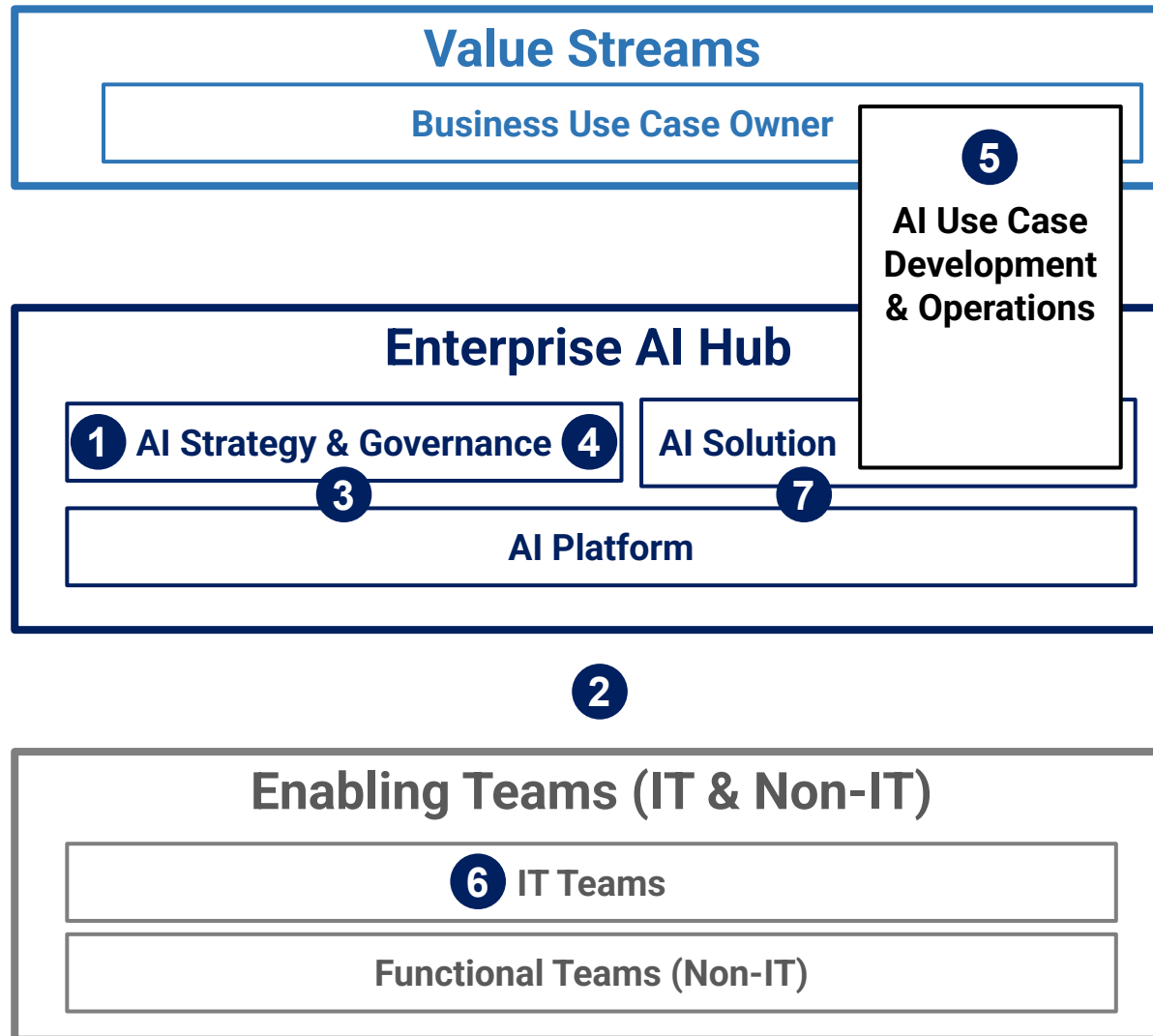
Enabling Teams support AI capability delivery and alignment to strategic, operational, and compliance goals

IT Teams: Own and deliver IT Services to enable AI Capabilities (e.g., Cloud & infrastructure, Security, Data & Insights)

Functional Teams (Non-IT): Enable and support AI capabilities with non-IT services (e.g., Finance, Legal)



The Enterprise AI Hub provides AI platform and solution capabilities to value streams, leveraging services from enabling teams



Key Interactions

1	Enterprise AI Hub Strategy & Governance defines and communicates enterprise AI strategy, standards, guardrails, and delivery guidance to Value Streams , Use Case , and Enabling Teams . Hub Strategy & Governance team will continuously drive strategy and seek feedback
2	Enterprise AI Hub aligns strategy, value, operational, and compliance processes with Enabling Teams and confirms delivery readiness
3	Enterprise AI Hub Strategy & Governance initiates development of the enterprise AI platform with Platform Team , establishing core capabilities, tooling, and foundational services
4	Enterprise AI Hub Strategy & Governance reviews and prioritizes use cases submitted by Value Streams
5	Enterprise AI Hub and Value Streams allocate resources and capacity to AI Use Case Development & Operations to deliver AI solutions, combining AI expertise with business context
6	IT Enabling Teams provision infrastructure, cloud, security, and data services while Functional teams provide financial and legal oversight to AI Platform and Business Use Case Owner Teams
7	AI Use Case Development & Operations leverage AI Platform team and capabilities to accelerate development and increase reusability



Within each team, smaller teams deliver AI capabilities with Solution Development and Operations shared between Hub and Value Streams

Value Streams

Provide business context, requirements and support AI use case development and operations

Business Use Case Owner

Business-led teams that define and steer AI use cases to solve specific business needs

AI Use Case Development & Operations

Designs, builds, and deploys and monitors, maintains, and supports AI solutions.

Teams comprised of Hub and/or Value Stream resources depending on Value Stream's AI proficiency

Enterprise AI Hub

Owens AI strategy, governance, platform, and expertise to scale and reuse AI

AI Strategy & Governance

Defines enterprise AI strategy, standards, and governance to ensure alignment with strategic priorities, architecture, and compliance

AI Platform

Builds and operates shared AI platform to enable teams to develop, deploy, and run AI solutions at scale

AI Platform Development & Operations

Knowledge Layer Team

Structures enterprise knowledge so AI can apply context

Agent Layer POD

Builds agent run time to allow agents to execute workflows

Agent Gov & Model Layer POD

Enforces guardrails for compliant agent actions

AI Platform Operations

Monitors, maintains, and supports AI platform

AI Solution

Provides centralized capacity to develop AI solutions and upskill resources in AI across Value Streams

Enabling Teams (IT & non-IT)

Support AI capability delivery and alignment to strategic, operational, and compliance goals

IT Enabling Teams

Own and deliver IT Services to enable AI Capabilities

Data & Analytics

Performance Management

OCIO

Cloud & Infra (CTO)

Security & Compliance

Functional Enabling Teams

Enable and support AI capabilities with non-IT services (e.g., Finance, Legal)



Agentic AI Operating Model High-Level RACI

Note: Leaders to be consulted and informed at appropriate times

Activities		Value Stream		Enterprise AI Hub			Enabling Teams	
		Business Use Case Owner	AI Use Case Development & Operations	AI Strategy & Governance	AI Platform	AI Solution	IT Enabling Teams	Functional Enabling Teams
1.0 AI Strategy & Tech Business Management	Define agentic AI strategy, goals, and priorities in alignment with business needs	C	C	R, A	C	C	I	I
	Oversee AI activities to ensure policy compliance, portfolio and value management, and talent upskilling	C	I	R, A	I	I	I	I
2.0 Solution Delivery & Management	Define use case and business requirements and prioritize investment	A	R	I	I	R	I	
	Design, develop, deploy and operate use case solutions	C	A, R	I	I	R	I	
	Provide development capacity for building use case solutions	C	C	A	C	R	I	
	Track value realization across business outcomes	A	I	R,A	I	I	I	
3.0 AI Platform	Manage and prioritize platform backlog aligned to business AI strategy and target outcomes	C	C	C	R,A	C	I	
	Develop and maintain platform capabilities, including support of use cases	C	I	I	R,A	I	I	
4.0 Data & Insights	Identify and enhance data products for use cases	I	C	I	I	I	R, A	
	Support integration testing	I	C	I	C	I	R, A	
5.0 Infrastructure, Operations, & Security	Setup cloud, infrastructure environments and authorize access	I	I	I	C	I	R, A	
	Define and implement AI security and cyber controls	I	I	I	C	I	R, A	
	Define and execute financial / audit / risk controls and legal compliance	I	C	C	I	I	I	R,A

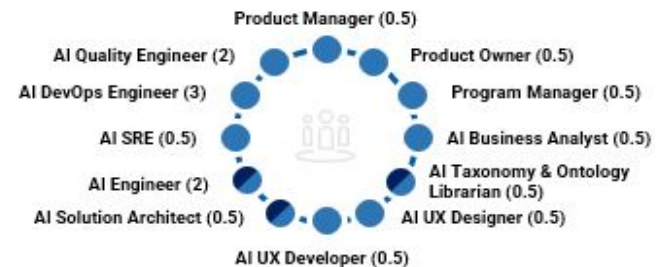
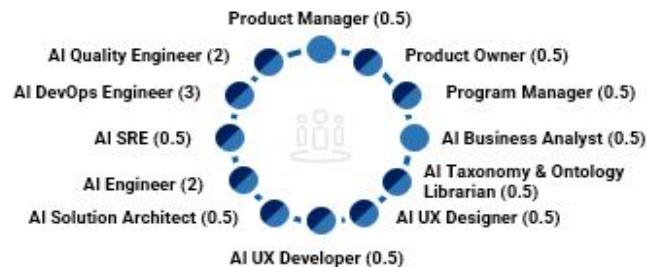


Use case delivery models depending on value stream AI skill maturity

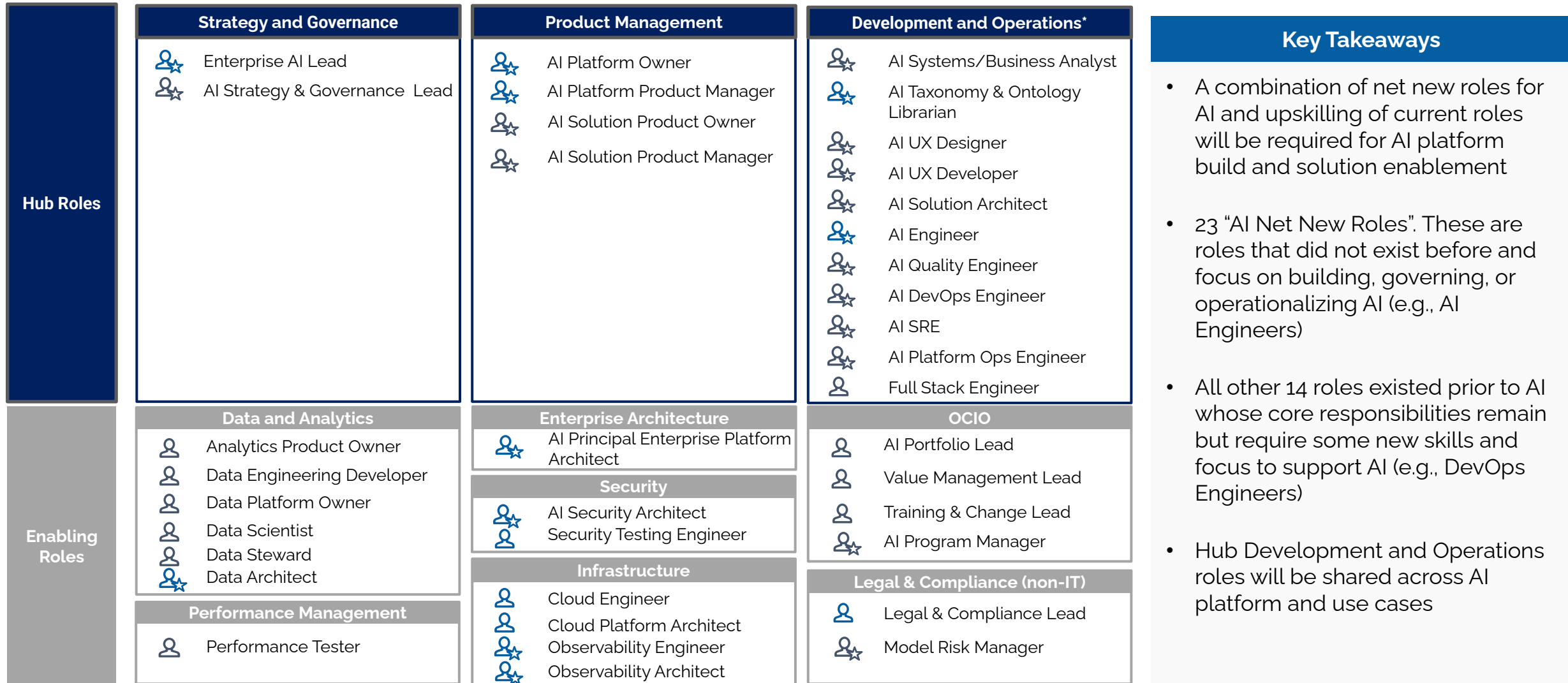
Legend Hub Value Stream

	White Glove Delivery <i>Centralized at the Hub</i>	Shared Delivery <i>Shared Hub and Value Stream</i>	Self Service <i>Value Stream owned with Hub support</i>
	Business Engagement Business Requirements Design Delivery Operations AI Platform	Business Engagement Business Requirements Design Delivery Operations AI Platform	Business Engagement Business Requirements Design Delivery Operations AI Platform
Value Stream AI Skill Maturity			
Hub	Owns design, delivery and operations of use cases	Shares design, delivery and operations of use cases	Provides technical guidance and support, as needed
Value Stream	Defines business requirements <i>Example: Legal</i>	Defines business requirements and shares design, delivery and operations of use cases <i>Example: Pharmacy Solutions</i>	Defines business requirements and owns design, delivery and operations of use cases <i>Example: eCommerce</i>

Illustrative Use Case Pod Composition



AI Hub and Enabling Roles



☆ AI New Roles

👤 MVP Roles (Medium)



All roles will need new skills in addition to understanding AI fundamentals and Responsible AI

From

Strategy

- Conceptual AI understanding
- Broad technology policy development

Architecture and Technical Design

- Service based architectures
- Static data modelling
- Limited semantic or contextual awareness

Platform and Product Management

- Feature-centric product roadmaps
- Limited transparency into AI behavior
- Manual prioritization and experimentation cycles

Development and Operations

- Traditional DevOps and ML Ops practices
- Manual model deployment and tuning
- Limited runtime intelligence or autonomy

To

Strategy

- AI Strategy
- AI Value Creation

Architecture and Technical Design

- Multi-Agent Orchestration Architecture
- Model Context Protocol
- Context Engineering
- Knowledge Engineering
- Taxonomy & Ontology Management

Platform and Product Management

- AI Experience Design
- Agent Lifecycle Management
- Agent Explainability &

Development and Operations

- Agent Ops
- LLM Ops
- Agent Monitoring
- Memory Management
- Multi-Agent Orchestration
- Model Context Protocol
- Context Engineering
- Knowledge Engineering
- Taxonomy & Ontology Management

From

Quality Engineering and Testing

- Deterministic test cases and scripted validation
- Limited insight into AI decision paths
- Post-release defect detection

Data and Analytics

- Structured data and BI-centric analytics
- Keyword-based search and reporting
- Manual insight generation

Infrastructure

- Static infrastructure provisioning
- Compute optimized for traditional workloads
- Limited support for dynamic AI workloads

Security

- Perimeter- and identity-focused security
- Reactive threat detection
- Limited AI-specific controls

Governance

- Traditional tech governance
- Static risk frameworks
- Limited operational enforcement

Technology Business Management

- Legacy technology value realization and cost management

To

Quality Engineering and Testing

- Agent Explainability &
- Agent Monitoring
- Reasoning & Acting (ReAct) Framework

Data and Analytics

- Vector Intelligence & Embeddings
- Natural Language Processing

Infrastructure

- Agent Monitoring
- Agent Lifecycle Management

Security

- Agent Security
- Agent Governance

Governance

- Agent Governance – Registry & Certification
- Agent Explainability &

Technology Business Management

- AI Project based value realization and cost management



Platform Journey from MVP to Sustain



Agentic AI: From Tools to Enterprise Platform

Agentic AI should be treated like cloud or data platforms — built once, governed centrally, and reused across the enterprise.

Key Shifts in the Marketplace



A surge in AI offerings is accelerating retailer adoption of Agentic AI across customer, merchandising, and operational decisions.



AI agents communicate directly with humans, other agents, workflows, and data. These communication pathways expand the attack surface.



Organizations are investing in Agentic AI platforms to deploy, manage, and govern agents at scale.



Growing recognition that AI solutions have a high failure rate and that technology alone is not the main constraint.

Why this Matters for Costco?



Agentic point solutions, when adopted in isolation, create agent sprawl that increases cost, duplication, and operational risk.



Disparate agents create security and compliance exposure by operating with inconsistent access controls, data permissions, and governance standards.



Platform accelerates delivery, enables reuse, governance, and cost discipline through shared agent capabilities, centralized controls, and economies of scale.



Continuous value tracking and business engagement is critical for agentic solutions as it ensures they deliver measurable business outcomes while controlling risk, performance, and spend.

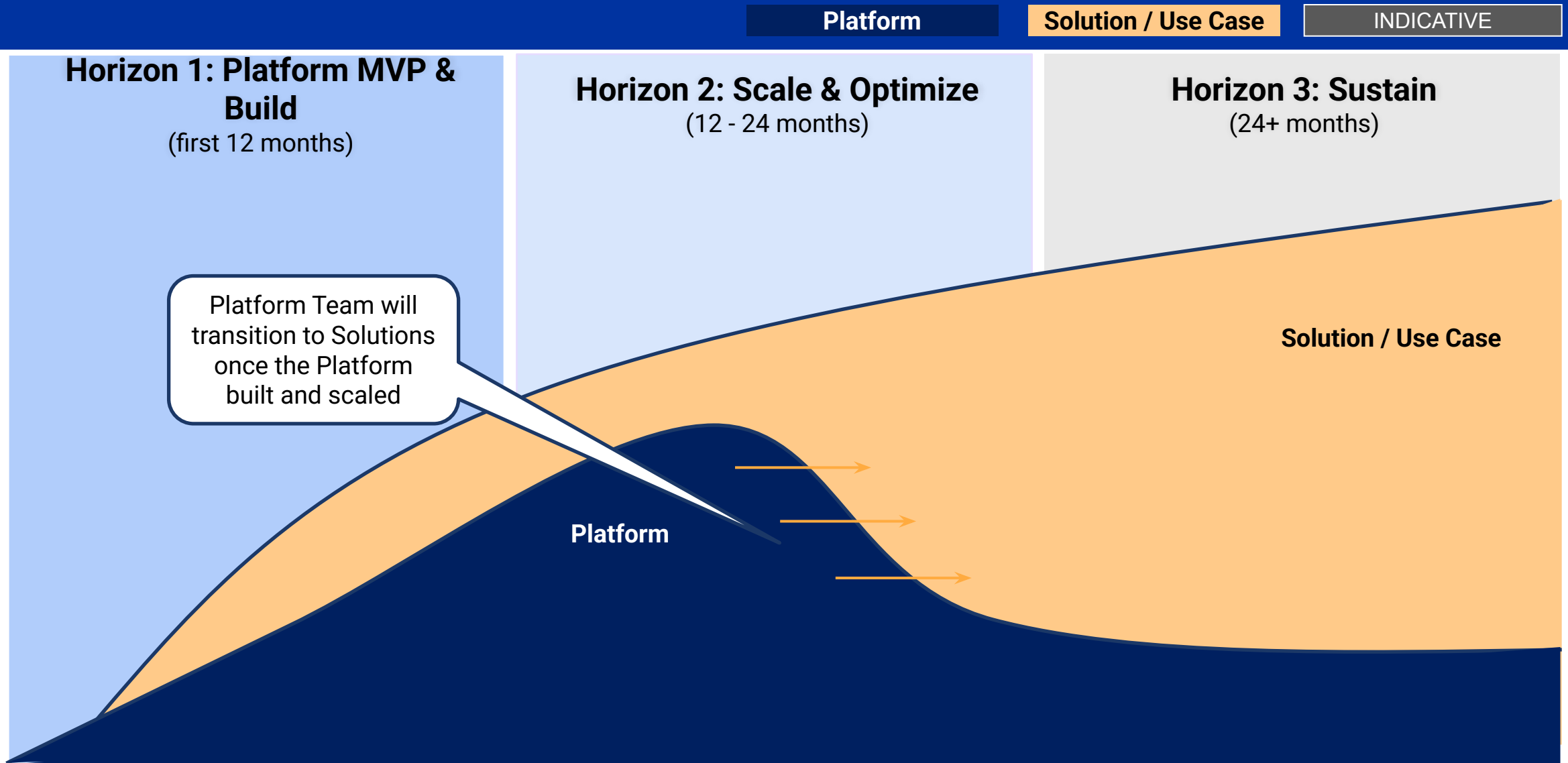


Agentic AI Journey across 3 Horizons

Horizon 1: Platform MVP & Build (first 12 months) <i>Prove value fast, establish agentic and architectural foundation</i>	Horizon 2: Scale & Optimize (12 - 24 months) <i>Enable multiple solutions, accelerate delivery, reduce friction</i>	Horizon 3: Sustain (24+ months) <i>Focus on stability, efficiency, and long-term value realization</i>
Enterprise AI Hub - AI Hub launched and fully operational - AI platform MVP is built and matured to be production-ready - Responsible AI and risk controls established	- AI Hub supports both platform and use case development - AI platform scales to support AI demand - Repeatable and scalable use case delivery - Stands up a dedicated Platform Ops team	- AI Hub scales down to smaller, long-term model - Enables Value Streams to be in a self-service model - AI platform shifts from platform development to efficient operations
Value Streams Teams 3-6 use case solutions mobilized - Establish upskilling program	4-6 Transformations Mobilized - Expand upskilling across all Value Streams	6-8 Transformations Mobilized - Value Streams can own use case development
Enabling Teams - Data and cloud foundation set, including FinOps measurement - AI portfolio and financial management processes established	- IT Enabling services optimized to scale - Value management processes established to inform governance decisions	- Mature cost control practices with ability to forecast cloud spend - AI portfolio actively optimized to maximize value realization - Continuous upskilling of AI Hub and Value Streams



Anticipated Effort & Ramp Up & Down for AI Platform and Solution Teams

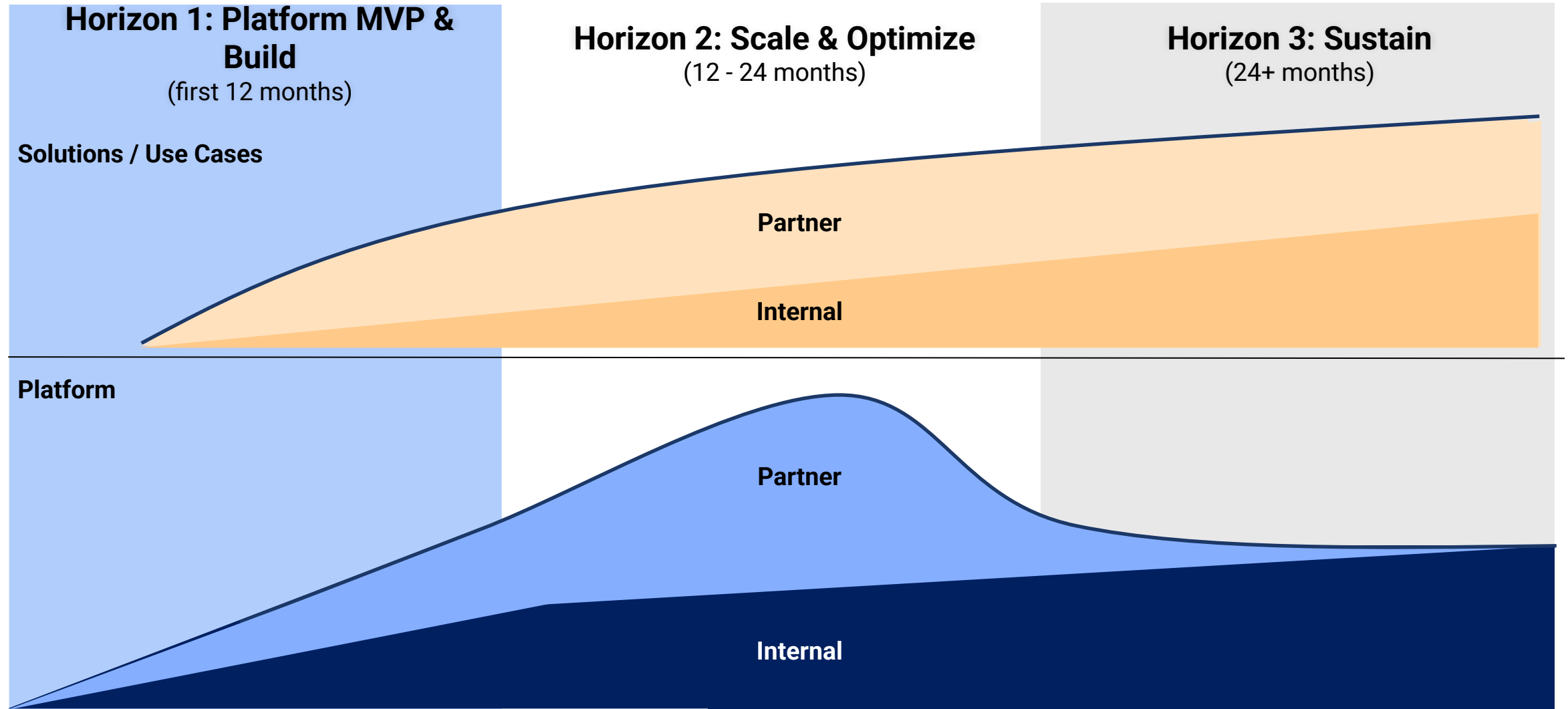


*Indicative view into anticipated effort required. Actual effort will vary based on the number and complexity of solutions / use cases, and enhancements for the platform.



Anticipated Effort & Ramp Up & Down for AI Platform and Solution Teams

Platform		Solution / Use Case	
Internal	Partner	Internal	Partner



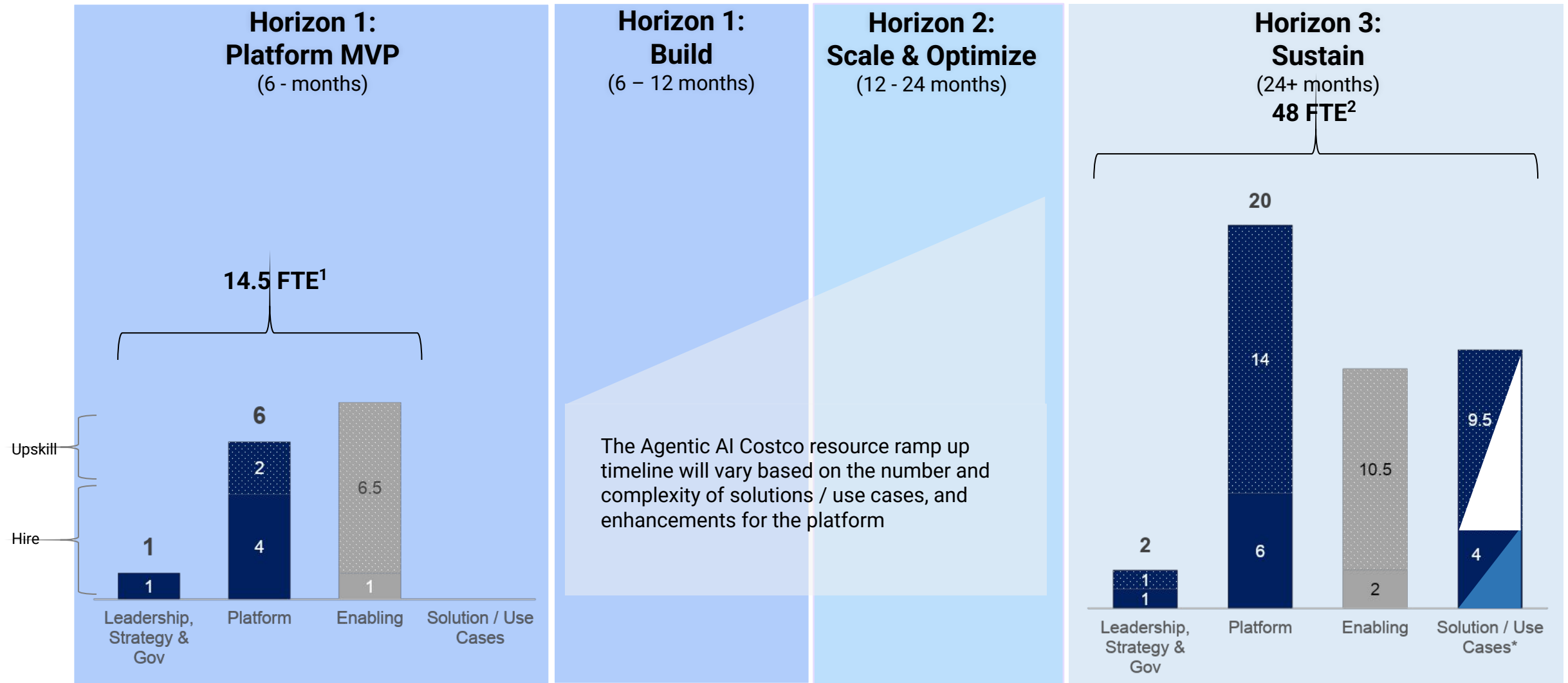
INDICATIVE

*Indicative view into anticipated effort required. Actual effort will vary based on the number and complexity of solutions / use cases, and enhancements for the platform.



Internal Agentic AI Resourcing Estimates through the Horizons

Platform MVP and Sustain



¹MVP: 17 roles of which 12 are full-time and 5 are part-time (0.5 FTE)

²Sustain: of the 54 roles 41 are full-time and 14 are part-time (0.5 FTE)

*Solution / Use Case teams in Horizon 3 assumes 1 use case team and will be a made up of AI Hub and Value Stream resources. For each additional use case an additional team will need to be deployed.

Horizon 3: “Sustain” Costco Resources – Organization View

Legend: **Bold:** Denotes People Manager

Team PODs

48 FTE

Enterprise AI Hub (35.5)

Enterprise AI Lead
AVP

1 FTE

AI Strategy &
Gov. Lead
Manager

16 FTE

AI Platform Owner
Director

AI Platform Product Mgr.
Manager

AI Principal Engineer
Principal

AI Systems / Bus. Analyst
Staff

Engineering Manager
Director

4 FTE

AI Platform
Operations Lead
Manager

- AI SRE
- AI Platform Ops Engineer
- AI Platform Ops Engineer

13.5 FTE

AI Solution Owner
Director

AI Sol Product Mgr (0.5)
Manager

AI Solution Architect
Director

AI Systems / Bus. Analyst
Staff

Engineering Manager
Director

11 FTE

Knowledge Layer (2 FTE)

- AI Senior Engineer
- AI Taxonomy and Ontology Librarian

Agent Layer (2 FTE)

- AI Senior Engineer
- Full-Stack Senior Engineer

Agent Gov. & Model (2 FTE)

- AI Senior Engineer
- Full-Stack Senior Engineer

Cross-POD (5 FTE)

- AI UX Designer
- AI UX Developer
- AI DevOps Engineer

- AI Quality Engineering Lead
- AI Quality Engineer

Solution POD (9 FTE)

- AI Senior Engineer
- AI Engineer
- AI Engineer
- Full-Stack Senior Engineer
- Full-Stack Engineer
- AI UX Designer
- AI UX Developer

- AI Quality Engineering Lead
- AI Quality Engineer

Enabling Teams (12.5)

Enterprise Architecture (1 FTE)

- Enterprise AI Principal Architect

Cloud & Infrastructure (1.5 FTE)

- Cloud Engineer
- Observability Engineer (0.5 FTE)

Data & Insights (4.5 FTE)

- Data Architect
- Data Engineer (1.5 FTE)
- Analytics Product Manager (0.5 FTE)
- Data Platform Owner (0.5 FTE)
- Data Scientist (0.5 FTE)
- Data Steward (0.5 FTE)

InfoSec (1 FTE)

- AI Security Architect (0.5 FTE)
- Security Testing Engineer (0.5 FTE)

Governance Risk & Compliance (1 FTE)

- Model Risk Manager

Legal (0.5 FTE)

- Legal Analyst (0.5 FTE)

Performance management (0.5 FTE)

- Performance Tester (0.5 FTE)

OCIO (3.5 FTE)

- AI Portfolio Lead (0.5 FTE)
- AI Program Manager (2)
- Value Management Lead (0.5 FTE)
- Training & Change Lead (0.5 FTE)



Costco Role Details (1 of 2)

#	Role	Level	Individual Contributor / People Manager	Staffing (Hire, Upskill)	Group	Sub-Group L2	Sourcing Location	MVP - Small	MVP - Medium	MVP - Large	Steady State	# of FTEs MVP	# of FTEs Steady State	Month Start
1	AI Platform Owner	Director	PM	Upskill	AI Hub	Platform Build	Onshore	Yes	Yes	Yes	Yes	1	1	0
2	AI Platform Product Manager	Manager	IC	Upskill	AI Hub	Platform Build	Onshore	Yes	Yes	Yes	Yes	1	1	0
3	AI Principal Engineer	Principal	IC	Hire	AI Hub	Platform Build	Onshore	Yes	Yes	Yes	Yes	1	1	1
4	AI Taxonomy and Ontology Librarian	Staff	IC	Hire	AI Hub	Platform Build	Onshore		Yes	Yes	Yes	1	1	3
5	AI Senior Engineer	Senior Staff	IC	Hire	AI Hub	Platform Build	Onshore		Yes	Yes	Yes	1	1	1
6	AI Senior Engineer	Senior Staff	IC	Hire	AI Hub	Platform Build	Onshore		Yes	Yes	Yes	1	1	2
7	AI Senior Engineer	Senior Staff	IC	Hire	AI Hub	Platform Build	Onshore			Yes	Yes	1	1	2
8	AI UX Designer	Senior Staff	IC	Hire	AI Hub	Platform Build	Onshore				Yes	1	1	2
9	AI SRE	Staff	IC	Upskill	AI Hub	Platform Operations	Offshore			Yes	Yes	1	1	3
10	Enterprise AI Lead	AVP	PM	Hire	AI Hub	Leadership	Onshore	Yes	Yes	Yes	Yes	1	1	0
11	AI Strategy & Governance Lead	Manager	IC	Upskill	AI Hub	Strategy & Governance	Onshore			Yes	Yes	1	1	2
12	AI DevOps Engineer	Staff	IC	Upskill	AI Hub	Platform Build	Onshore				Yes	1	1	8
13	AI Systems / Business Analyst	Staff	IC	Upskill	AI Hub	Platform Build	Onshore				Yes	1	1	8
14	Engineering Manager	Director	PM	Upskill	AI Hub	Platform Build	Offshore				Yes	1	1	6
15	Full-Stack Senior Engineer	Senior Staff	IC	Upskill	AI Hub	Platform Build	Onshore				Yes	1	1	6
16	Full-Stack Engineer	Staff	IC	Upskill	AI Hub	Platform Build	Offshore				Yes	1	1	8
17	AI UX Developer	Staff	IC	Upskill	AI Hub	Platform Build	Offshore				Yes	1	1	9
18	AI Quality Engineering Lead	Senior Staff	IC	Upskill	AI Hub	Platform Build	Offshore				Yes	1	1	10
19	AI Quality Engineer	Staff	IC	Upskill	AI Hub	Platform Build	Offshore				Yes	1	1	10
20	AI Platform Operations Lead	Manager	IC	Upskill	AI Hub	Platform Operations	Onshore				Yes	1	1	11
21	AI Platform Operations Engineer	Staff	IC	Upskill	AI Hub	Platform Operations	Onshore				Yes	1	1	12
22	AI Platform Operations Engineer	Staff	IC	Upskill	AI Hub	Platform Operations	Offshore				Yes	1	1	12
23	AI Solution Owner	Director	PM	Upskill	AI Hub	Solution	Onshore				Yes	1	1	4
24	AI Solution Product Manager	Manager	IC	Upskill	AI Hub	Solution	Onshore				Yes	0.5	0.5	4
25	AI Solution Architect	Director	IC	Hire	AI Hub	Solution	Onshore				Yes	1	1	5
26	AI Systems / Business Analyst	Staff	IC	Upskill	AI Hub	Solution	Onshore				Yes	1	1	5
27	Engineering Manager	Director	PM	Upskill	AI Hub	Solution	Onshore				Yes	1	1	5
28	AI Senior Engineer	Senior Staff	IC	Hire	AI Hub	Solution	Onshore				Yes	1	1	5
29	AI Engineer	Staff	IC	Hire	AI Hub	Solution	Onshore				Yes	1	1	6
30	AI Engineer	Staff	IC	Hire	AI Hub	Solution	Offshore				Yes	1	1	6



Costco Role Details – (2 of 2)

#	Role	Level	Individual Contributor / People Manager	Staffing (Hire, Upskill)	Group	Sub-Group L2	Sourcing Location	MVP - Small	MVP - Medium	MVP - Large	Steady State	# of FTEs MVP	# of FTEs Steady State	Month Start
31	Full-Stack Senior Engineer	Senior Staff	IC	Upskill	AI Hub	Solution	Onshore				Yes	1	1	6
32	Full-Stack Engineer	Staff	IC	Upskill	AI Hub	Solution	Offshore				Yes	1	1	6
33	AI Quality Engineering Lead	Senior Staff	IC	Upskill	AI Hub	Solution	Offshore				Yes	1	1	7
34	AI Quality Engineer	Staff	IC	Upskill	AI Hub	Solution	Offshore				Yes	1	1	7
35	AI UX Designer	Senior Staff	IC	Upskill	AI Hub	Solution	Onshore				Yes	1	1	4
36	AI UX Developer	Staff	IC	Upskill	AI Hub	Solution	Offshore				Yes	1	1	5
37	Enterprise AI Principal Architect	Principal	IC	Hire	Enabling	EA	Onshore	Yes	Yes	Yes	Yes	1	1	0
38	AI Platform Program Manager	Manager	IC	Upskill	Enabling	VMO	Onshore			Yes	Yes	1	1	1
39	Cloud Engineer	Staff	IC	Upskill	Enabling	Cloud & Infra	Offshore	Yes	Yes	Yes	No	1	0	1
40	Data Engineer	Staff	IC	Upskill	Enabling	Data & Insights	Offshore				Yes	1	1	2
41	AI Security Architect	Director	IC	Upskill	Enabling	InfoSec	Onshore	Yes	Yes	Yes	No	0.5	0	1
42	Security Testing Engineer	Staff	IC	Upskill	Enabling	InfoSec	Onshore	Yes	Yes	Yes	Yes	0.5	0.5	3
43	Model Risk Manager	Senior Staff	IC	Hire	Enabling	Legal & Compliance	Onshore				Yes	1	1	7
44	Observability Engineer	Senior Staff	IC	Upskill	Enabling	Observability	Onshore	Yes	Yes	Yes	Yes	0.5	0.5	2
45	AI Program Manager	Manager	IC	Upskill	Enabling	VMO	Onshore				Yes	1	1	4
46	Analytics Product Manager	Manager	IC	Upskill	Enabling	Data & Insights	Offshore				Yes	0.5	0.5	7
47	Data Engineering Developer	Staff	IC	Upskill	Enabling	Data & Insights	Offshore				Yes	0.5	0.5	7
48	Data Platform Owner	Director	IC	Upskill	Enabling	Data & Insights	Offshore				Yes	0.5	0.5	5
49	Data Scientist	Director	IC	Upskill	Enabling	Data & Insights	Offshore				Yes	0.5	0.5	6
50	Data Steward	Senior Staff	IC	Upskill	Enabling	Data & Insights	Offshore				Yes	0.5	0.5	5
51	AI Portfolio Lead	Director	IC	Upskill	Enabling	OCIO	Onshore				Yes	0.5	0.5	9
52	Value Management Lead	Manager	IC	Upskill	Enabling	OCIO	Onshore				Yes	0.5	0.5	8
53	Training & Change Lead	Director	IC	Upskill	Enabling	OCIO	Onshore				Yes	0.5	0.5	0
54	Performance Tester	Senior Staff	IC	Upskill	Enabling	Performance Mgmt.	Offshore				Yes	0.5	0.5	5
55	Cloud Engineer	Staff	IC	Upskill	Enabling	Cloud & Infra	Onshore	Yes	Yes	Yes	Yes	1	1	1
56	Data Architect	Director	IC	Upskill	Enabling	Data & Insights	Onshore	Yes	Yes	Yes	Yes	0.5	0.5	2
57	Cloud Architect	Director	IC	Upskill	Enabling	Cloud & Infra	Onshore	Yes	Yes	Yes	No	1	0	1
58	Legal Analyst	Senior Staff	IC	Upskill	Enabling	Legal & Compliance	Onshore	Yes	Yes	Yes	Yes	0.5	0.5	1
59	Observability Architect	Director	IC	Upskill	Enabling	Observability	Onshore	Yes	Yes	Yes	No	1	0	1



Shoring Considerations

There is opportunity for technology services to be delivered from a nearshore or offshore location

	Onshore	Nearshore	Offshore
Description	Entire team stays local, close to the Business	Move to a location which is easily accessible (may be abroad or within same geographic location)	Entire team moves to a remote location
Best suited for:	Highly specialized talent / domain knowledge that is scarce even in large financial centers / high-cost locations Activities where interactions are in-person with both external and internal clients Highly proprietary activities that the business doesn't feel comfortable developing in remote locations	More stable / mature products and processes Low complexity activities where the onshore team still need to maintain an element of control and rigor Low Business impact activities Processes that demand real-time responsiveness and time zone continuity	Very stable systems and mature processes Most interactions can be handled remotely via phone or email Typically, standard / commodity / highly available skills Situations that do not require immediate turnaround requirements
Not suited for:	High-volume, repetitive tasks that can be standardized and automated Cost-sensitive operations Functions that do not require in-person collaboration or proximity to the business	Specialized or proprietary work that requires niche expertise Activities with significant in-person collaboration needs	Business-critical functions requiring immediate turnaround or real-time decision-making Complex, high-touch processes that require frequent collaboration Activities involving sensitive intellectual property or regulatory restrictions



Rationale for Shoring Recommendations

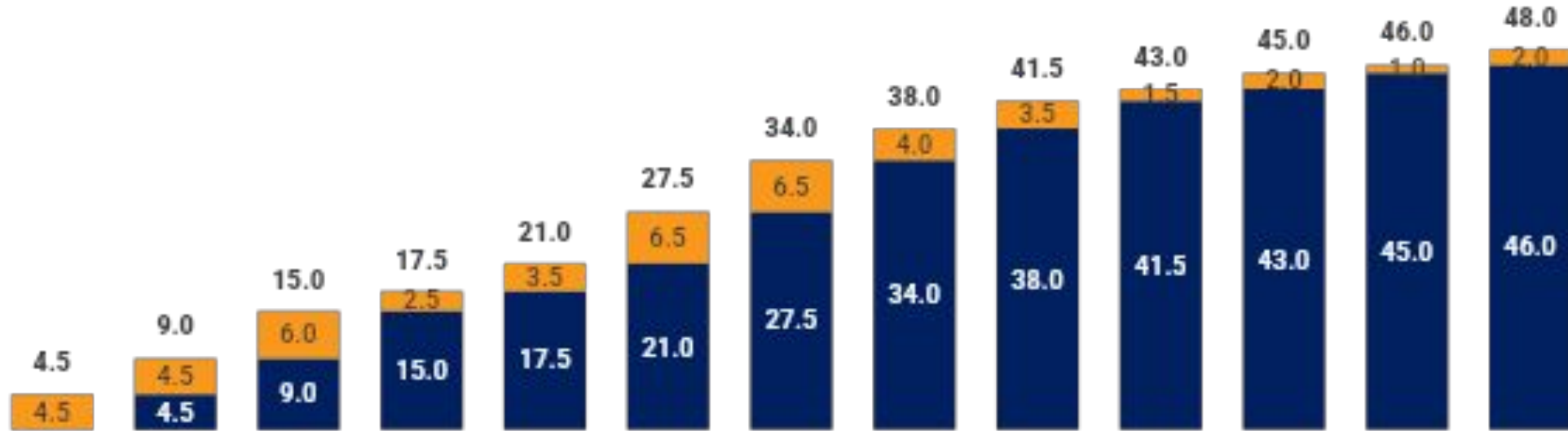
Role	Level	FTE	IC / People Manager	Group	Sub Group	Sub-Group L2	Capacity Provided By	Rationale
AI Engineer	Staff	1	IC	AI Hub	Solution	Solution	Engineering	Portion of the engineering capacity can be delivered from offshore and allows to start building an AI capabilities offshore. Engineering Manager identified due to offshore regulations typically require a local people manager.
Full-Stack Engineer	Staff	1	IC	AI Hub	Solution	Solution	Engineering	
Full-Stack Engineer	Staff	1	IC	AI Hub	Platform	Platform Build	Engineering	
Engineering Manager	Director	1	People Manager	AI Hub	Platform	Platform Build	Engineering	
AI UX Developer	Staff	1	IC	AI Hub	Platform	Platform Build	UX	UX Developers can be from offshore as there are mature capabilities offshore.
AI UX Developer	Staff	1	IC	AI Hub	Solution	Solution	UX	UX Designer however will be onshore.
AI Quality Engineering Lead	Senior Staff	1	IC	AI Hub	Platform	Platform Build	QE	Quality Engineering is process-driven, repeatable, scalable, and tool-enabled which makes it well suitable for offshoring.
AI Quality Engineer	Staff	1	IC	AI Hub	Platform	Platform Build	QE	
AI Quality Engineering Lead	Senior Staff	1	IC	AI Hub	Solution	Solution	QE	
AI Quality Engineer	Staff	1	IC	AI Hub	Solution	Solution	QE	
AI Platform Operations Engineer	Staff	1	IC	AI Hub	Platform	Platform Ops	IT Operations	Operations capabilities are good candidates for offshoring which also gives possibility to provide after hours support.
AI SRE	Staff	1	IC	AI Hub	Platform	Platform Ops	CTO	
Cloud Engineer	Staff	1	IC	Enabling	Enabling	Cloud & Infra	Cloud & Infra	Cloud, Infrastructure, and Data Services are good candidates for offshoring as the as the services are well defined and typically have clear SLAs.
Data Engineer	Staff	1	IC	Enabling	Enabling	Data & Insights	Data & Insights	
Analytics Product Manager	Manager	0.5	IC	Enabling	Enabling	Data & Insights	Data & Insights	
Data Engineering Developer	Staff	0.5	IC	Enabling	Enabling	Data & Insights	Data & Insights	
Data Platform Owner	Director	0.5	IC	Enabling	Enabling	Data & Insights	Data & Insights	
Data Scientist	Director	0.5	IC	Enabling	Enabling	Data & Insights	Data & Insights	
Data Steward	Senior Staff	0.5	IC	Enabling	Enabling	Data & Insights	Data & Insights	
Performance Tester	Senior Staff	0.5	IC	Enabling	Enabling	Perf. Mgmt.	Perf. Mgmt.	



Ramp Up Plan for AI Hub, Use Case Solution, and Enabling Teams

Newly Onboarded

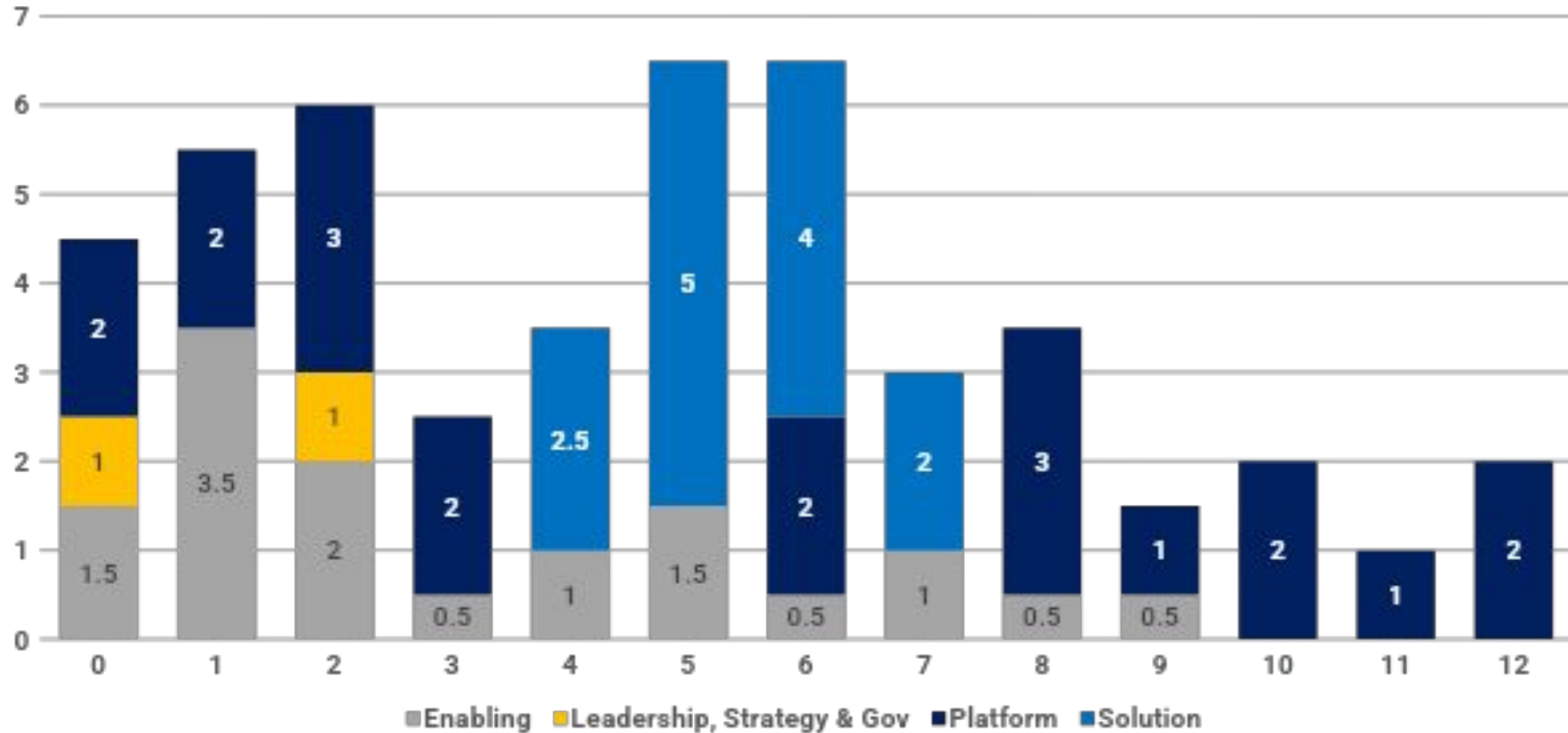
Previously Onboarded



Months	0	1	2	3	4	5	6	7	8	9	10	11	12	Total
Platform	2	2	3	2			2		3	1	2	1	2	20
Enabling	1.5	2.5	2	0.5	1	1.5	0.5	2	0.5	0.5				12.5
Leadership, Strategy & Gov	1		1											2
Solution					2.5	5	4	2						13.5
Total	4.5	4.5	6	2.5	3.5	6.5	6.5	4	3.5	1.5	2	1	2	48



Monthly Onboarding by Team



Horizon 1: MVP & Build



Horizon 1: Minimum Viable Platform (MVP)

What does “Viable” Look Like?

Horizon 1:
Platform
MVP & Build

Horizon 2:
Scale &
Optimize

Horizon 3:
Sustain

AI Platform MVP Objective: Establish a flexible foundation to scale Agentic AI use cases and test architecture decisions.

MVP Platform Focus



Environment Build

- ✓ Provision the Cloud environment and buildout of foundational components



Proofs of Technology / Concept

- ✓ Conduct Proof of Technology (POT) to select the right tools for platform



Foundational Agentic Capabilities

- ✓ AI Gateway
- ✓ Evaluation & Framework
- ✓ Knowledge Layer
- ✓ Utility Agents (Intent and Disambiguation)

Excludes



Solution / Use Case Development

- ❑ Excludes solution or use-case development, prioritizing the delivery of core platform services, shared utilities, and governance



Enterprise Rollout

- ❑ Excludes enterprise-wide deployment across value streams



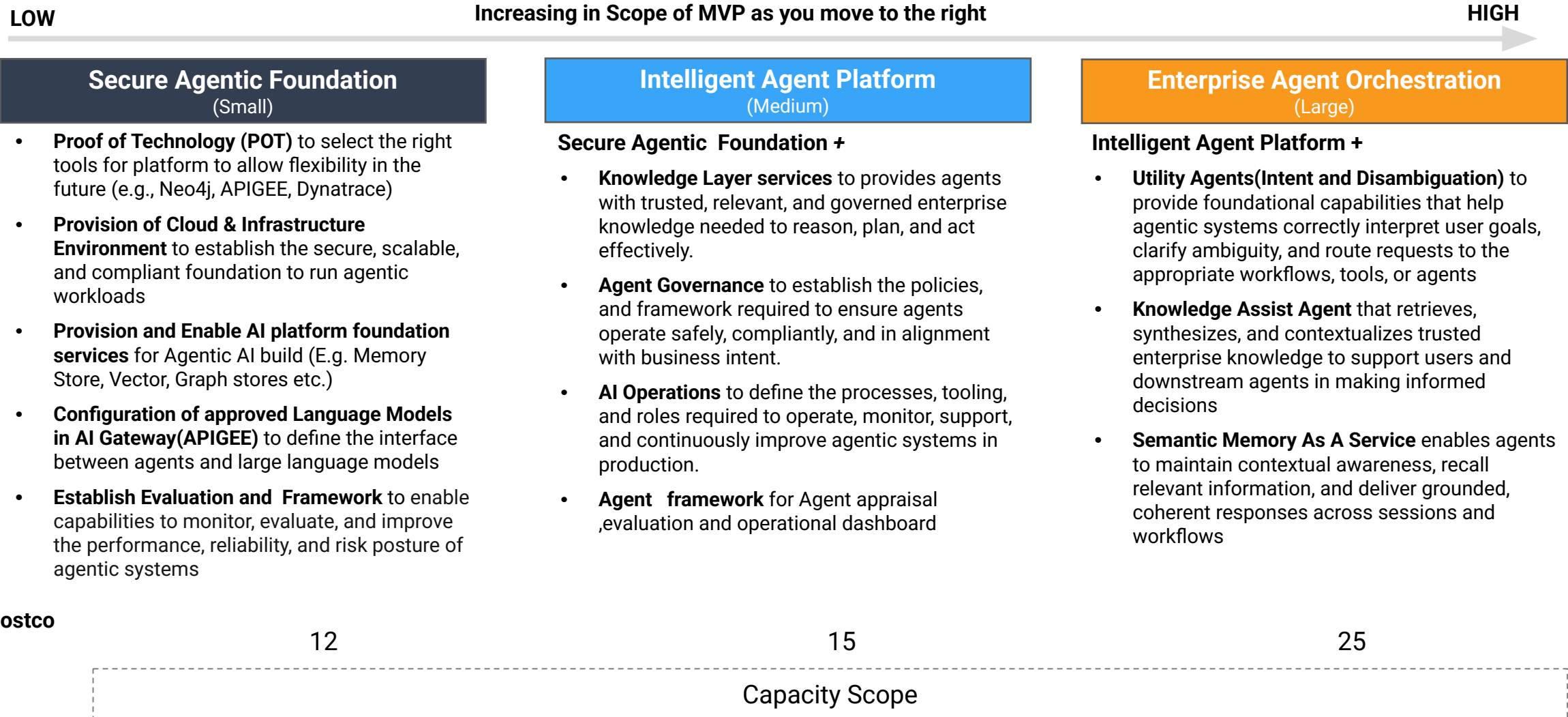
Value Stream Change Management

- ❑ Excludes business end-user change management, training and communications activities within value streams



AI Platform MVP Options & Scope

Decision Point: How quickly you want to move?



Agentic AI Platform Capabilities

“Enterprise Agent Orchestration” Option

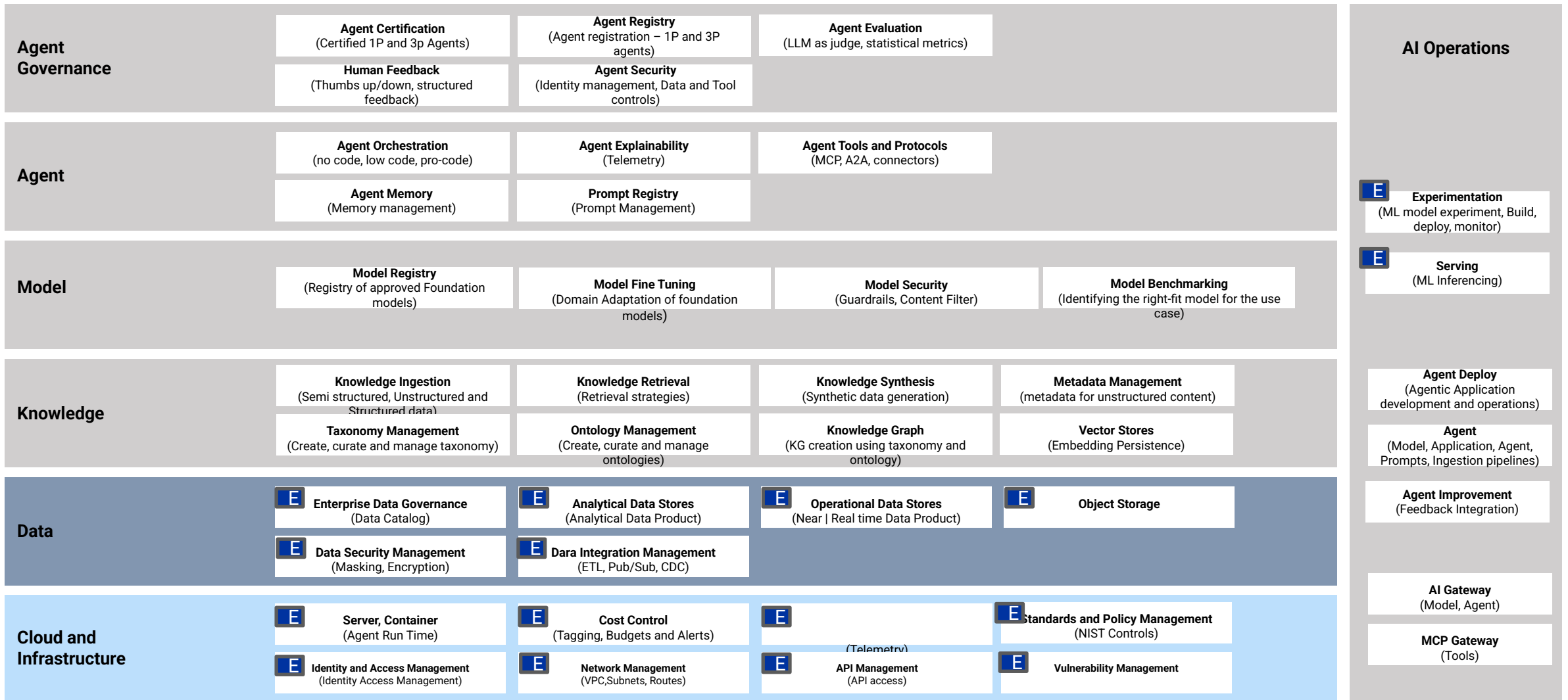
Capability

EXISTING CAPABILITY

CLOUD & INFRA

PLATFORM

DATA



Agentic AI Platform Capabilities

“Enterprise Agent Orchestration” (Medium)

Capability

EXISTING CAPABILITY

CLOUD & INFRA

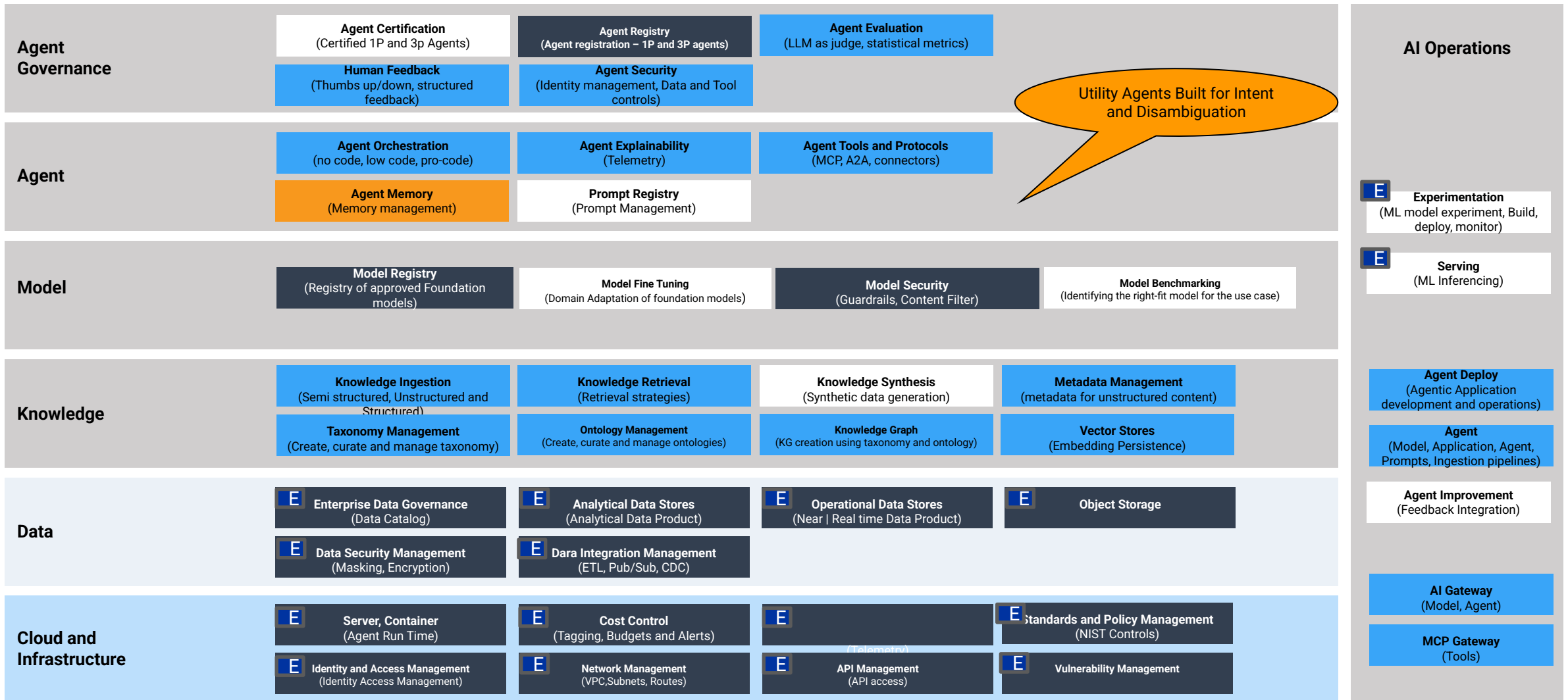
PLATFORM

DATA

Secure Agentic Foundation

Intelligent Agent Platform

Enterprise Agent Orchestration



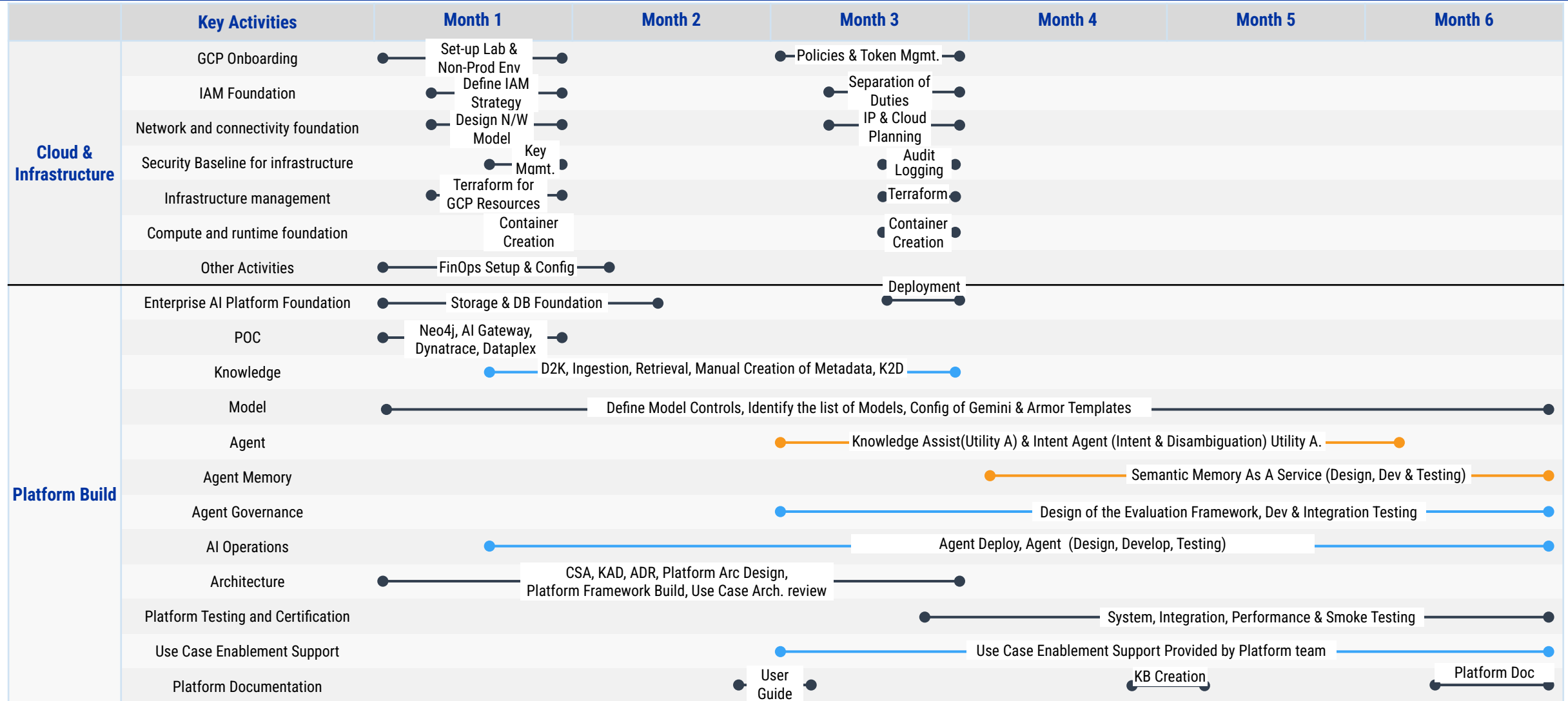
Agentic AI Platform Foundation Build Roadmap

Legend

Secure Agentic Foundation (Small)

Intelligent Agent Platform (Medium)

Enterprise Agent Orchestration (Large)



Horizon 1: MVP Costco Resources – Organization View

Intelligent Agent Platform (Medium) Option

Legend: **Bold:** Denotes People Manager

Team PODs

14.5 FTE

Enterprise AI Hub (7)

Enterprise AI Lead
AVP

6 FTE

AI Platform Owner
Director

AI Platform Product Mgr.
Manager

AI Principal Engineer
Principal

Knowledge Layer (2 FTE)

- AI Senior Engineer
- AI Taxonomy and Ontology Librarian

Agent Layer

Agent Gov. & Model (1 FTE)

- AI Senior Engineer

Solution POD

Cross-POD

Enabling Teams (7.5)

Enterprise Architecture (1 FTE)

- Enterprise AI Principal Architect

Cloud & Infrastructure (4.5 FTE)

- Cloud Architect
- Cloud Engineer (2 FTE)
- Observability Architect
- Observability Engineer (0.5 FTE)

Data & Insights (0.5 FTE)

- Data Architect (0.5 FTE)

InfoSec (1 FTE)

- AI Security Architect (0.5 FTE)
- Security Testing Engineer (0.5 FTE)

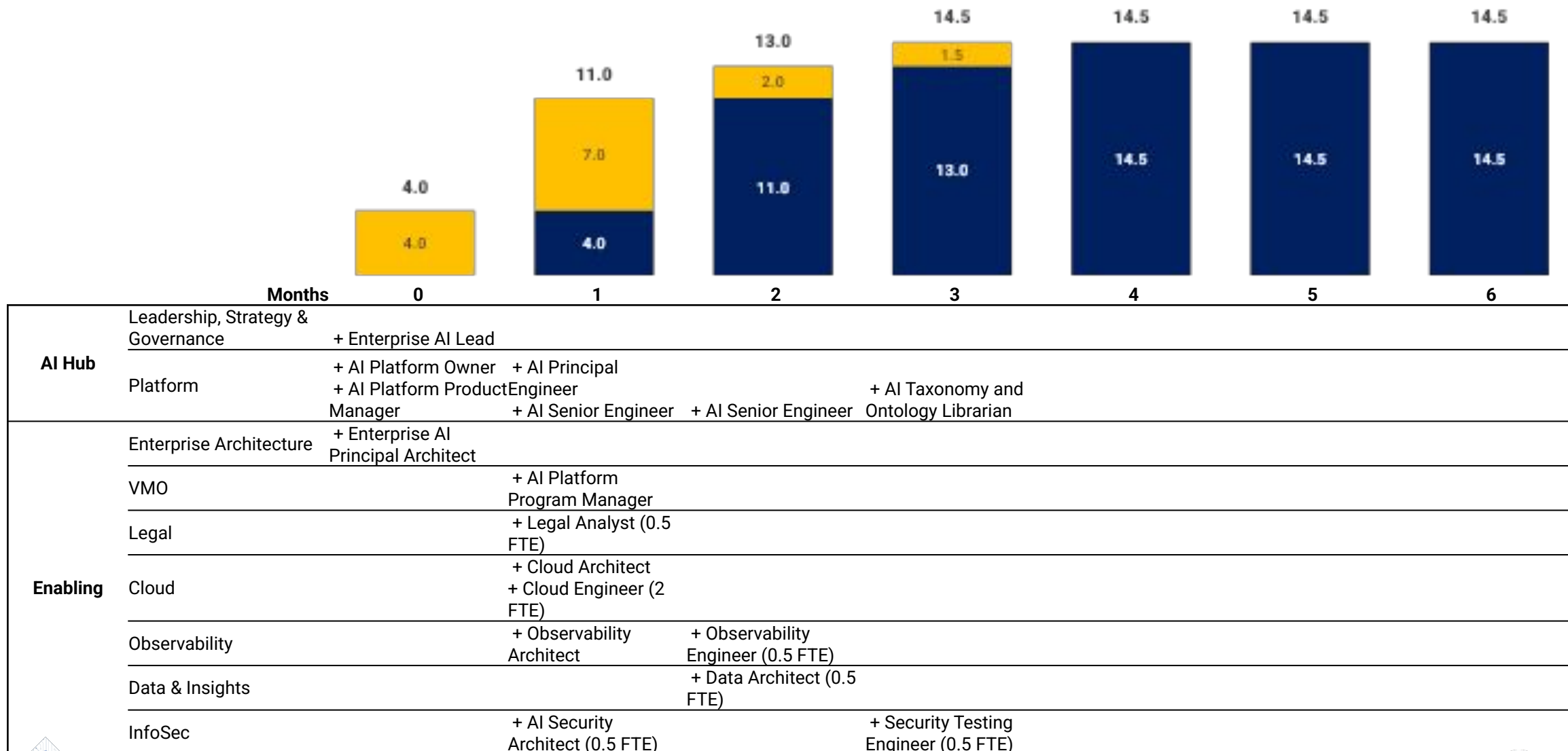
Legal (0.5 FTE)

- Legal Analyst (0.5 FTE)



Platform MVP Ramp up Plan for Costco Internal Resources

Intelligent Agent Platform (Medium) Option



Note:

- Only includes roles needed for AI Platform MVP development
- Onboarding will occur in the beginning of each month

Newly Onboarded

Previously Onboarded



Horizon 1: MVP Costco Resources – Organization View

Enterprise Agent Orchestration (Large) Option

Legend: **Bold:** Denotes People Manager

Team PODs

18.5 FTE

Enterprise AI Hub (10)

Enterprise AI Lead
AVP

1 FTE

AI Strategy &
Gov. Lead
Manager

8 FTE

AI Platform Owner
Director

AI Platform Product Mgr.
Manager

AI Principal Engineer
Principal

AI SRE
Staff

Knowledge Layer (2 FTE)

- AI Senior Engineer
- AI Taxonomy and Ontology Librarian

Agent Layer (1 FTE)

- AI Senior Engineer

Agent Gov. & Model (1 FTE)

- AI Senior Engineer

Cross-POD

Enabling Teams (8.5)

Enterprise Architecture (1 FTE)

- Enterprise AI Principal Architect

Cloud & Infrastructure (4.5 FTE)

- Cloud Architect
- Cloud Engineer (2 FTE)
- Observability Architect
- Observability Engineer (0.5 FTE)

Data & Insights (0.5 FTE)

- Data Architect (0.5 FTE)

InfoSec (1 FTE)

- AI Security Architect (0.5 FTE)
- Security Testing Engineer (0.5 FTE)

Legal (0.5 FTE)

- Legal Analyst (0.5 FTE)

OCIO (1 FTE)

- AI Program Manager

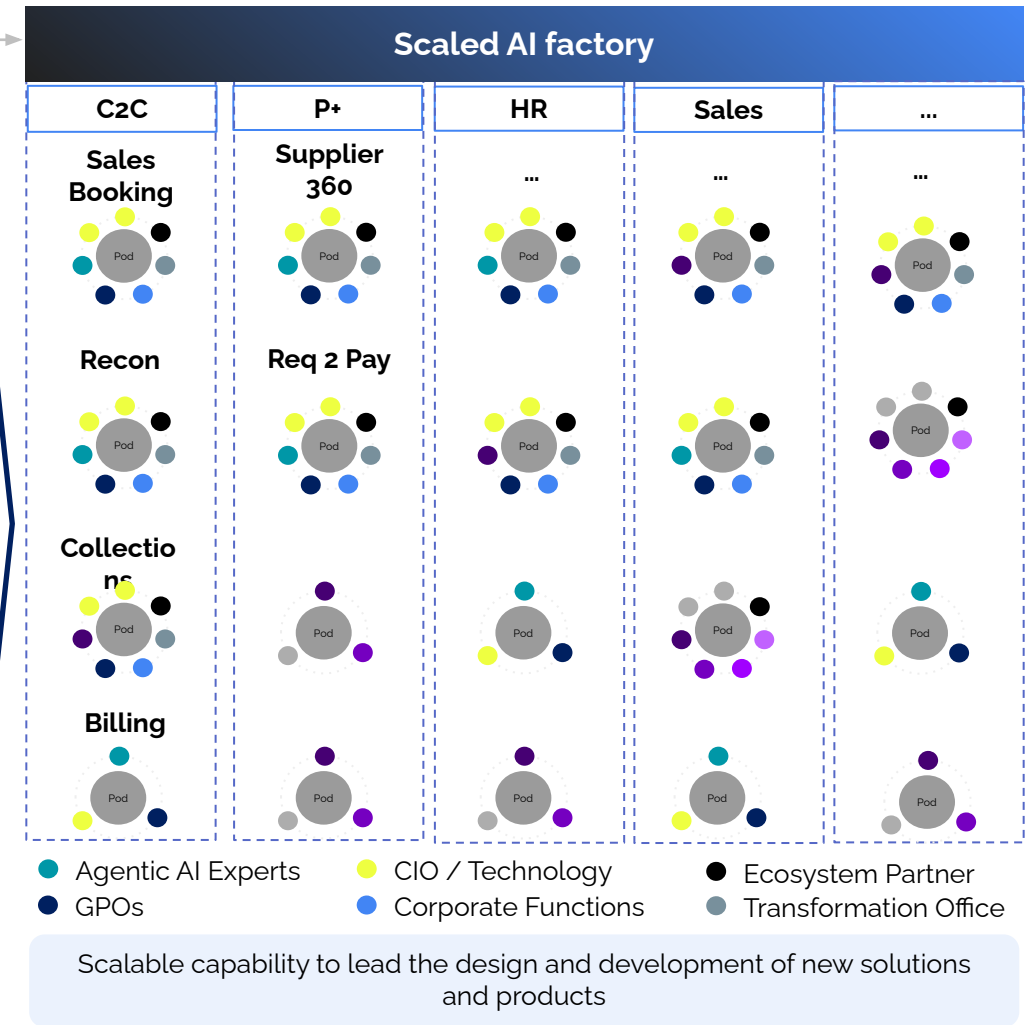
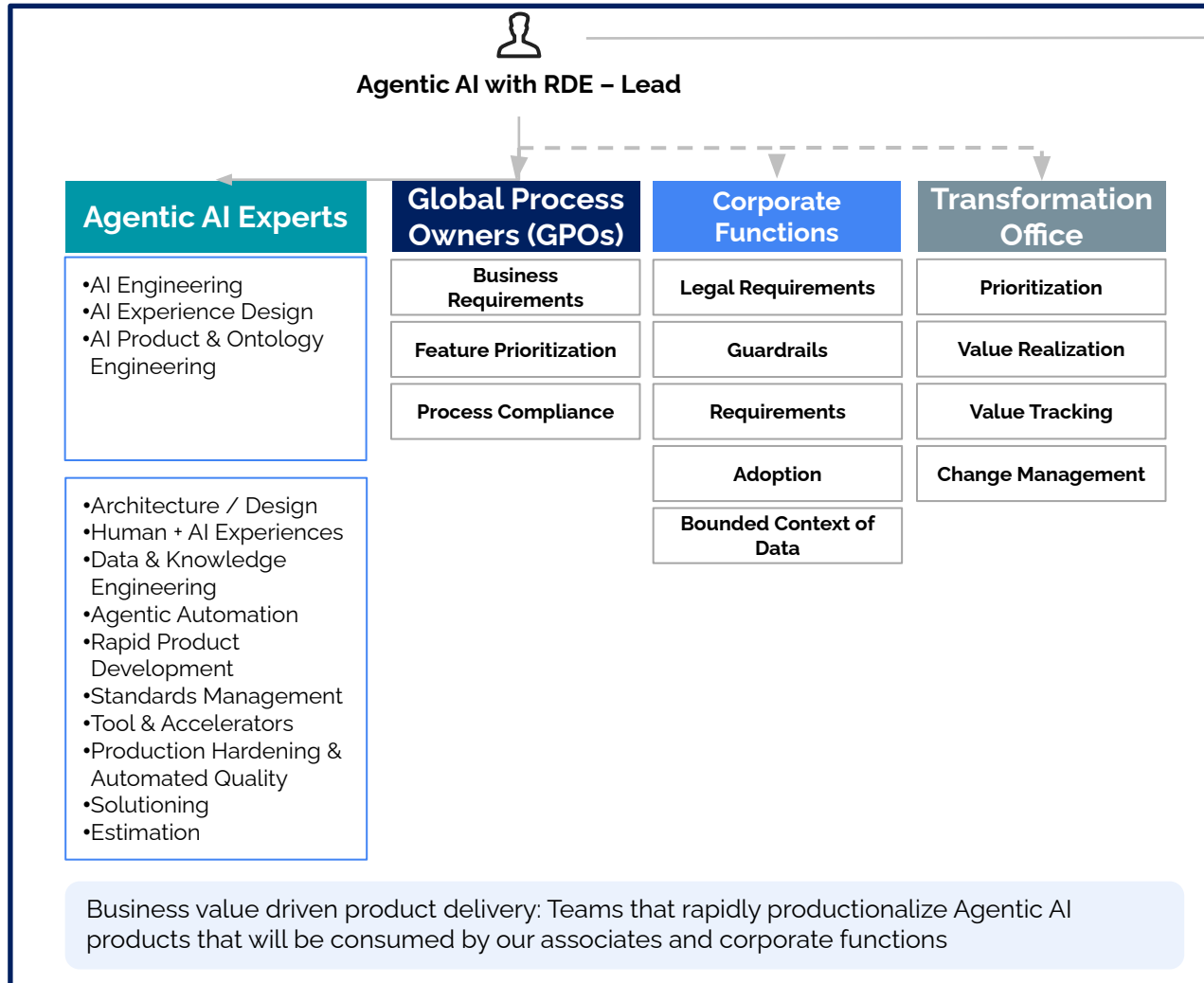


APPENDIX:

Example Factory Delivery Model



Example Factory Model for Repeatability, and Scale



APPENDIX: Role Details



All roles will need new skills in addition to understanding AI fundamentals and Responsible AI

From

Strategy

- Conceptual AI understanding
- Broad technology policy development

Architecture and Technical Design

- Service based architectures
- Static data modelling
- Limited semantic or contextual awareness

Platform and Product Management

- Feature-centric product roadmaps
- Limited transparency into AI behavior
- Manual prioritization and experimentation cycles

Development and Operations

- Traditional DevOps and MLOps practices
- Manual model deployment and tuning
- Limited runtime intelligence or autonomy

To

Strategy

- AI Strategy
- AI Value Creation

Architecture and Technical Design

- Multi-Agent Orchestration Architecture
- Model Context Protocol
- Context Engineering
- Knowledge Engineering
- Taxonomy & Ontology Management

Platform and Product Management

- AI Experience Design
- Agent Lifecycle Management
- Agent Explainability &

Development and Operations

- AgentOps
- LLMOps
- Agent Monitoring
- Memory Management
- Multi-Agent Orchestration
- Model Context Protocol
- Context Engineering
- Knowledge Engineering
- Taxonomy & Ontology Management

From

Quality Engineering and Testing

- Deterministic test cases and scripted validation
- Limited insight into AI decision paths
- Post-release defect detection

Data and Analytics

- Structured data and BI-centric analytics
- Keyword-based search and reporting
- Manual insight generation

Infrastructure

- Static infrastructure provisioning
- Compute optimized for traditional workloads
- Limited support for dynamic AI workloads

Security

- Perimeter- and identity-focused security
- Reactive threat detection
- Limited AI-specific controls

Governance

- Traditional tech governance
- Static risk frameworks
- Limited operational enforcement

Technology Business Management

- Legacy technology value realization and cost management

To

Quality Engineering and Testing

- Agent Explainability &
- Agent Monitoring
- Reasoning & Acting (ReAct) Framework

Data and Analytics

- Vector Intelligence & Embeddings
- Natural Language Processing

Infrastructure

- Agent Monitoring
- Agent Lifecycle Management

Security

- Agent Security
- Agent Governance

Governance

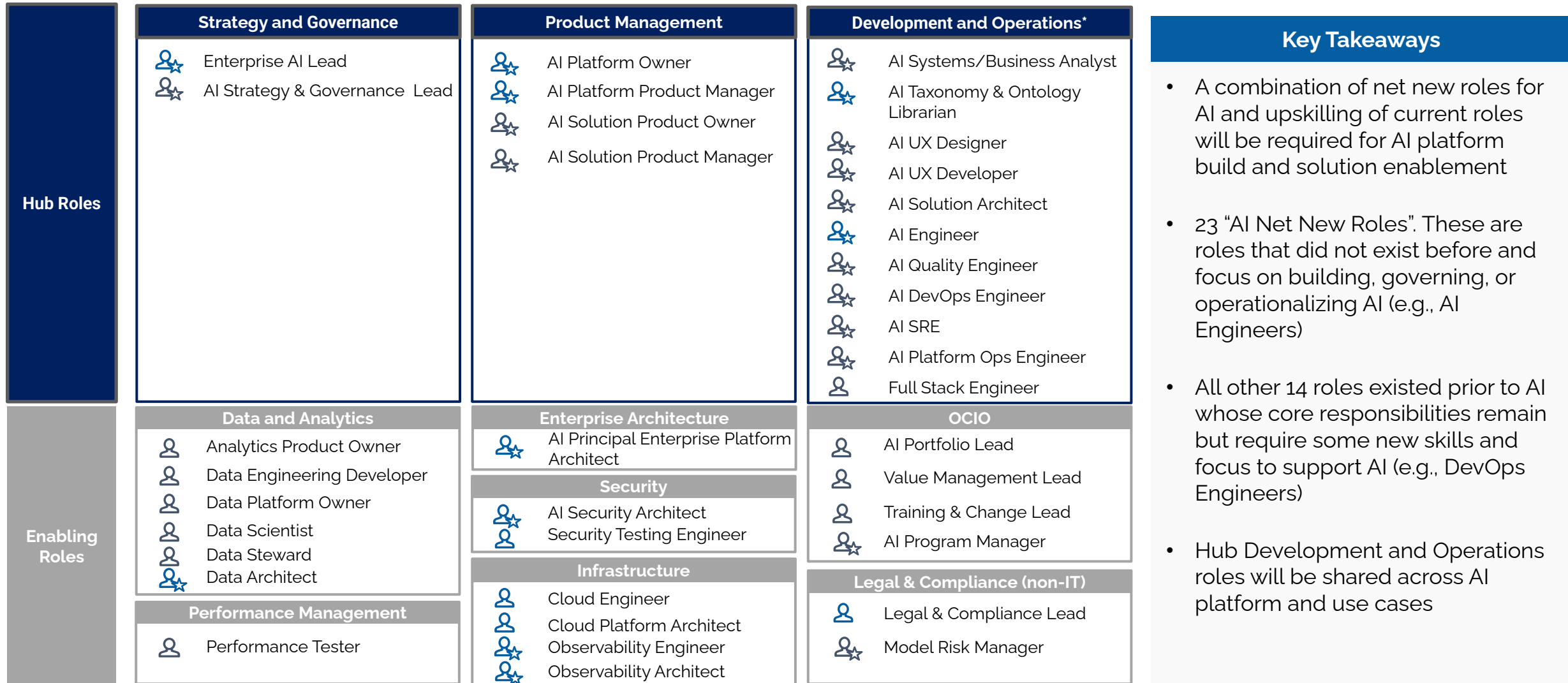
- Agent Governance – Registry & Certification
- Agent Explainability &

Technology Business Management

- AI Project based value realization and cost management



AI Hub and Enabling Roles



☆ AI New Roles

👤 MVP Roles (Medium)



Skills required to implement AI capabilities and align to chosen Tech Stack

Functional AI Skills

Guide AI platform and solution design, management and business alignment

Foundations & Core AI Concepts

- AI, GenAI, Agentic AI Fundamentals
- Natural Language Processing
- Responsible AI

Design & Knowledge Modeling

- AI Experience Design
- Knowledge Engineering
- Taxonomy and Ontology Management

Context, Memory & Intelligence

- Context Engineering
- Memory Management
- Vector Intelligence and Embeddings
- Model Context Protocol

Reasoning & Agent Behavior

- Prompt Engineering
- Reasoning & Acting (ReAct) Framework
- Multi-Agent Orchestration Architecture

Development and Operations

- LLM Ops
- Agent Ops
- Agent and Model Lifecycle Management

& Monitoring

- Agent Monitoring
- Agent Explainability and

Security, Governance & Trust

- Agent Security
- Agent Governance – Registry Certification

Technical AI Skills

Enable the hands-on building, integration, and running of AI models, agents, and platforms

Cloud Foundation

- GCP Storage and Compute Stack
- Alloy DB
- Redis Cache/Cloud memory store

Data Governance & Catalog

- Dataplex

Model & Agent Development

- Vertex AI
- Gemini Enterprise
- Google ADK
- Python

Retrieval & Memory

- Vector Search
- Neo4J
- mem0
- Langmem

API Gateway & Traffic Management

- APIGEE
- Kong

LLM Routing / Model Access Abstraction

- LiteLLM

Agent Observability

- Grafana
- Dynatrace

Evaluation & Quality

- RAGAS/DeepEvals
- Trulens

AI Security & Risk Controls

- Model Armor

LLM Tracing & Monitoring

- Arize



Roles Cards Index for Net New Roles

Role Name	Slide Link
Enterprise AI Lead	Link
AI Strategy and Governance Lead	Link
AI Principal Enterprise Architect	Link
AI Platform Owner	Link
AI Platform Product Manager	Link
AI Solution Product Owner	Link
AI Solution Product Manager	Link
AI Program Manager	Link
AI Systems/Business Analyst	Link
AI Taxonomy & Ontology Librarian	Link

Role Name	Slide Link
AI UX Designer	Link
AI UX Developer	Link
AI Solution Architect	Link
AI Engineer	Link
AI Quality Engineer	Link
AI DevOps Engineer	Link
AI Platform Operations Engineer	Link
AI SRE	Link
AI Security Architect	Link
Model Risk Manager	Link
Observability Engineer	Link



The role card provides a snapshot into the profile of net new roles for AI at Costco

The role cards are designed at a general level; however, specific responsibilities, functions, and skill requirements should be customized based on team alignment and individual role needs.

Position summary for the role

Job duties and essential functions

Recommended functional skills

AI Engineer

**Position Summary**

The AI Engineer designs, builds, and operates AI systems that enable scalable, secure, and reliable AI capabilities across the enterprise. This role focuses on developing and integrating AI-driven services, including agent-based workflows, model integrations, orchestration, and observability, to support both shared platforms and business solutions. The AI Engineer applies strong software engineering practices to ensure AI solutions are production-ready, governed, and aligned with enterprise standards. This role partners closely with product managers, architects, security, and operations teams to deliver AI capabilities that support business outcomes and responsible AI adoption.

**Job Duties and Essential Functions**

- Designs, develops, and maintains AI systems and services, including agent workflows, orchestration, and model integrations
- Implements prompt, context, memory, and tool management to enable consistent and reusable AI behavior
- Builds and supports telemetry, logging, and observability for AI workloads in production.

**Functional Skills**

- Agentic AI system design and orchestration
- Generative AI and LLM application engineering
- Prompt, memory, and tool management
- RAG, vector search, and knowledge engineering

- AI observability, evaluation, and quality assurance
- Responsible AI, governance, and security

**Technical Skills**

- Generative AI development with Gemini Enterprise on GCP
- Prompt, context, memory, and tool orchestration
- Retrieval-Augmented Generation (RAG) and vector search with Gemini Enterprise
- Knowledge engineering with graphs, embeddings, and Neo4j

- Integrates AI capabilities with enterprise systems, data platforms, and APIs.
- Ensures AI solutions meet scalability, reliability, security, and governance requirements.
- Collaborates with product, architecture, and operations teams to deliver roadmap capabilities.
- Supports testing, debugging, and continuous improvement of AI services

- Cloud-native AI architecture on GCP
- API-first enterprise integration (Apigee, Kong)
- Model, prompt, and agent lifecycle management (LLMOps)
- AI observability and evaluation (Trulens)



Costco's technology stack should also be considered for role acquisition and development depending on team assigned

- Alloy DB
- APIGEE
- Arize
- Dataplex
- Dynatrace
- GCP Storage and Compute Stack
- Gemini Enterprise
- Google ADK
- Grafana
- Kong
- Langmem
- LiteLLM

- mem0
- Model Armor
- Neo4J
- Python
- RAGAS/DeepEval s
- Redis
- Cache/Cloud memory store
- Trulens
- Vector Search
- Vertex AI

51

Enterprise AI Lead



Position Summary

The Enterprise AI Lead defines and drives the enterprise AI vision and strategy to enable scalable, secure, and responsible AI capabilities across the organization. The role oversees AI platform development and integration, governance and standards, and the prioritization and delivery of use cases to ensure solutions are production-ready, reusable, and aligned to business objectives. The Enterprise AI Lead partners with product, architecture, security, cloud, data, and operations teams to coordinate roadmaps, streamline execution, and accelerate adoption of enterprise-grade AI solutions.



Job Duties and Essential Functions

- Leads enterprise AI strategy, roadmap creation, and alignment with organizational priorities.
- Oversees AI platform evolution, ensuring scalability, reliability, compliance, and reusability.
- Establishes Responsible AI principles, governance models, and quality standards.
- Manages relationships with AI vendors from strategy and selection to governance and value realization
- Prioritizes and oversees AI initiatives across value streams to maximize business outcomes.
- Guides integration of AI systems with enterprise platforms, data layers, and cloud environments.
- Partners with cross-functional teams to drive adoption, change management, and capability uplift.



Functional Skills

- AI strategy and enterprise vision
- AI governance
- AI value creation
- Enterprise AI operating model design
- Organizational leadership and stakeholder management
- Project and program management for AI initiatives
- Responsible AI, risk, and compliance oversight
- Roadmap planning



Technical Skills

- AI platform, data, and knowledge architecture (e.g. Google Cloud Stack)
- Stakeholder communication and storytelling tools(e.g., Google Slides)
- Reporting and modelling platforms / tools
- Data, semantic, and knowledge architecture tools (e.g., Dataplex, Neo4J)



AI Strategy and Governance Lead



Position Summary

The AI Strategy & Governance Lead defines and drives the enterprise AI vision, strategy, and governance framework to ensure scalable, secure, responsible, and business-aligned AI adoption. This role aligns AI initiatives with corporate priorities and enterprise architecture while establishing policies, guardrails, and operating models that enable compliant and value-driven AI at scale. The leader partners cross-functionally with business, architecture, data, cloud, security, legal, risk, and product teams to shape AI strategy, prioritize initiatives, enforce governance standards, manage risk.



Job Duties and Essential Functions

- Define and lead the enterprise AI vision, strategy, and multi-year roadmap aligned to business objectives.
- Establish and maintain AI governance frameworks, policies, decision rights, and Responsible AI standards.
- Identify, prioritize, and oversee high-value AI initiatives across the enterprise portfolio.
- Implement lifecycle governance, risk controls, and assurance processes across AI development and deployment.
- Partner cross-functionally to ensure production readiness, regulatory compliance, and enterprise alignment.
- Monitor AI portfolio performance, risk posture, compliance adherence, and value realization.



Functional Skills

- AI strategy and vision
- AI governance
- AI value opportunity identification and prioritization
- AI change management strategy
- Cross-enterprise AI roadmap
- Enterprise AI operating model design
- Responsible and scalable AI adoption strategy
- Strategic alignment across business and technology functions



Technical Skills

- AI platform, data, and knowledge architecture (e.g. Google Cloud Stack)
- Stakeholder communication and storytelling tools (e.g., Google Slides)
- Reporting and modelling tools



AI Principal Enterprise Architect



Position Summary

The AI Principal Enterprise Architect defines the enterprise-wide AI target architecture, standards, and integration patterns needed to build scalable, secure, and interoperable AI capabilities. The role ensures the AI platform—including models, agents, data flows, orchestration layers, governance enablers, and cloud components—is architected for reliability, reuse, compliance, and alignment with enterprise technology strategies. The AI Enterprise Platform Architect partners closely with engineering, data, security, cloud, product, and governance teams to harmonize the platform ecosystem and enable seamless end-to-end AI solution delivery.



Job Duties and Essential Functions

- Defines target state AI platform architecture, including components, integration patterns, and interoperability standards.
- Guides implementation of platform capabilities including model access, routing, orchestration, memory, retrieval, and .
- Establishes technical standards, guardrails, reusable components, and platform-wide patterns for engineering teams.
- Ensures platform alignment with cloud, data, infrastructure, and security strategies across the enterprise.
- Reviews solution architectures for compliance, scalability, safety, and adherence to platform direction.
- Supports performance, reliability, resilience, and optimization assessments across AI workloads.



Functional Skills

- Agent and model lifecycle architecture
- AI platform reference architecture design
- AI security, governance, and compliance
- AI technical leadership & advisory
- Enterprise AI architecture
- Integration Pattern Design (APIs, Events, Services)
- Multi-Agent orchestration architecture
- Responsible AI Scalability, resilience & performance engineering



Technical Skills

- Enterprise AI platform architecture (e.g., Vertex AI, Gemini Enterprise)
- Agentic AI and multi-agent orchestration tools (e.g., Google ADK)
- Model, agent, and workflow lifecycle tools (e.g., Vertex AI)
- Retrieval, memory, and context architecture tools (e.g., Vector Search, Redis Cache / Cloud Memory store)
- Data, semantic, and knowledge architecture tools (e.g., Dataplex, Neo4J)
- AI Security systems (e.g., Model Armor)



AI Platform Owner



Position Summary

The AI Platform Owner owns the enterprise AI platform vision, target state, and multi-year roadmap, ensuring the platform evolves into a coherent, reusable, scalable, and secure set of capabilities that enable high-quality AI development across the organization. The role prioritizes platform capabilities such as model access, routing, orchestration, retrieval, memory, , security, and compliance, while driving alignment with enterprise strategy, architecture, and governance requirements. The AI Platform Owner partners closely with engineering, data, cloud, product, security, and governance teams—along with value streams and solution teams—to translate requirements, coordinate execution, and ensure the platform continuously delivers long-term business value.



Job Duties and Essential Functions

- Owns the platform vision, target architecture, and strategic roadmap aligned to enterprise AI strategy.
- Defines, prioritizes, and oversees delivery of core platform capabilities across model, agent, data, and orchestration layers.
- Establishes platform standards, reusable patterns, golden paths, and lifecycle management practices.
- Coordinates with engineering, cloud, data, security, and governance owners to ensure integrated, high-quality execution.
- Engages with value streams and solution teams to gather requirements, drive adoption, and support reuse.
- Monitors platform health, performance, risk posture, and continuous improvement opportunities.



Functional Skills

- Agent and model lifecycle management
- Enterprise AI platform vision and target-state definition
- Multi-year AI platform roadmap ownership
- Core platform capability prioritization
- AI , security, and compliance capability ownership
- Reusable and scalable AI platform design principles
- AI strategy alignment
- AI Platform governance and certification enablement



Technical Skills

- AI platform capability domain knowledge (e.g., Vertex AI, Gemini Enterprise)
- Platform support systems for AgentOps and LLMOps (e.g., Google ADK)
- Cloud-native AI platform services (e.g., GCP)
- Platform scalability, resilience, and cost control awareness (GCP Storage and Compute Stack)



AI Platform Product Manager



Position Summary

The AI Platform Product Manager owns the definition, prioritization, and lifecycle of AI platform capabilities, ensuring the platform evolves into a coherent, reusable, and scalable product ecosystem that meets the needs of AI developers, operators, and governance stakeholders. The role translates platform strategy into clear requirements, backlogs, and roadmaps, shaping capabilities such as model access, orchestration, retrieval, , and compliance enablement. Delivers platform features that are enterprise-grade, secure, and aligned to long-term value, partnering closely with engineering, architecture, data, cloud, security, and governance teams to ensure coordinated execution and consistent adoption across the organization.



Job Duties and Essential Functions

- Owns AI platform capability roadmap, defining and prioritizing features aligned with platform strategy.
- Translates user and enterprise needs into clear requirements, stories, and structured backlogs.
- Works with engineering, architecture, and infrastructure teams to deliver scalable, secure platform capabilities.
- Ensures platform features support Responsible AI, governance, , and compliance expectations.
- Partners with value streams and solution teams to capture requirements and drive platform adoption.
- Tracks capability performance, user satisfaction, platform reuse, and continuous improvement opportunities.



Functional Skills

- AI platform capability prioritization and planning
- AI platform value definition and adoption
- AI product management
- AI risk, dependency and scoping
- AI strategy
- Developer and operator alignment
- Lifecycle management of AI platforms
- Platform governance and Responsible AI
- Requirement gathering & stakeholder alignment
- Road-mapping and release planning



Technical Skills

- Platform roadmap and backlog management tools
- Capability definition and requirement documentation (e.g., Confluence, Google Docs)
- Platform dependency and integration mapping (e.g., Miro, Lucidchart)
- Usage, adoption, and value tracking tools (e.g., Jira, Power BI)
- Decision logging and architectural traceability tools (e.g., Confluence)
- Stakeholder communication tools (e.g., Google Slides)



AI Solution Product Owner



Position Summary

The AI Solution Product Owner owns the vision, strategic direction, and multi-horizon roadmap for AI solutions, ensuring they align with enterprise priorities, platform capabilities, and governance expectations. The role defines what AI solutions should exist, why they matter, and how they evolve over time, providing clear direction for delivery teams and stakeholders. The AI Solution Product Owner partners closely with business leaders, platform owners, architects, governance, and product managers to shape solution intent, guide prioritization decisions, and ensure AI solutions deliver sustained business value, responsible usage, and long-term adoption



Job Duties and Essential Functions

- Owns the AI solution vision, strategic intent, and roadmap aligned to enterprise and domain objectives.
- Defines solution scope, capability boundaries, and evolution paths across models, agents, workflows, and integrations.
- Establishes solution priorities based on business value, risk, platform readiness, and governance considerations.
- Provides strategic direction to Product Managers and delivery teams to guide execution and sequencing.
- Ensures AI solutions align with platform standards, architectural principles, and Responsible AI expectations.
- Partners with business, platform, architecture, and governance leaders to drive adoption and long-term value realization.



Functional Skills

- AI strategy
- AI capability and scope framing
- AI experience design
- Agent and model lifecycle management
- Business value driven prioritization
- Cross-stakeholder coordination
- Solution strategy and roadmap planning
- Responsible AI



Technical Skills

- AI solution design and delivery platforms (Vertex AI, Gemini Enterprise)
- Prompt, context, and interaction design for AI solutions (LLMs, agent-based systems)
- Retrieval-augmented generation (RAG) and knowledge integration tools (e.g., Vertex Vector Search, Neo4j)
- AI solution , quality, and evaluation tools (e.g., TruLens, RAGAS)



AI Solution Product Manager



Position Summary

The AI Solution Product Manager owns the vision, roadmap, and delivery of AI-enabled products and solutions, ensuring they are aligned with business outcomes, user needs, and enterprise AI strategy. The role shapes solution capabilities end-to-end—from defining product direction and prioritizing features to guiding development teams on AI-driven functionality, user experience, and responsible usage. It delivers measurable value through clear roadmaps, high-quality releases, and continuous iteration, partnering closely with business stakeholders, engineering, data, design, platform, and governance teams to ensure adoption, compliance, and impact.



Job Duties and Essential Functions

- Owns AI solution capability roadmap, defining and prioritizing features aligned with platform strategy.
- Defines and prioritizes solution requirements, user stories, and acceptance criteria.
- Guides engineering and data teams on integrating models, agents, retrieval, and workflows into the product.
- Ensures solutions are secure, responsible, reliable, and compliant with enterprise governance.
- Partners with business, UX, and platform teams to validate features, refine designs, and drive adoption.
- Measures product performance using KPIs, feedback loops, and continuous improvement mechanisms.



Functional Skills

- AI product vision and strategy definition
- Outcome-driven roadmap and backlog prioritization
- AI feature and capability framing
- User-centered and AI-aware product discovery
- Value hypothesis definition and validation
- Cross-functional product leadership and alignment
- Responsible AI and governance-aware product design
- Release planning and incremental value delivery
- Adoption, feedback, and continuous product iteration



Technical Skills

- Solution roadmap and backlog management tools
- Capability requirement documentation (e.g., Confluence, Google Docs)
- Decision logging and architectural traceability tools (e.g., Confluence)
- Usage, adoption, and value tracking tools (e.g., Jira, Power BI)
- Solution dependency and integration mapping tools (e.g., Miro, Lucidchart)
- Stakeholder communication and storytelling tools (e.g., Google Slides)



AI Program Manager



Position Summary

The AI Program Manager owns the coordinated execution of AI initiatives across teams, ensuring predictable, scalable delivery of the AI roadmap by managing timelines, dependencies, risks, and cross-functional alignment. The role orchestrates end-to-end program execution—from planning through release—while establishing delivery rhythms, tracking progress, and ensuring AI solutions meet business outcomes, governance expectations, and user needs. It delivers structured program plans, milestone visibility, and value-focused execution, partnering closely with product owners, engineering teams, data and cloud teams, governance functions, and business stakeholders to remove blockers and enable effective adoption.



Job Duties and Essential Functions

- Leads planning and coordinated execution of AI initiatives across multiple teams and workstreams.
- Manages program timelines, inter-team dependencies, risks, issues, and delivery milestones.
- Ensures alignment with AI strategy, governance requirements, and enterprise delivery standards.
- Partners with engineering, data, cloud, and product teams to maintain delivery quality and unblock progress.
- Drives stakeholder communication, reporting, and cross-functional alignment throughout execution.
- Tracks program outcomes, value realization, readiness, and continuous improvement opportunities.



Functional Skills

- AI program planning and roadmap execution
- AI delivery governance and milestone coordination
- AI project governance
- Cross-initiative dependency and risk management
- Release planning coordination for AI solutions
- Responsible AI
- Technical leadership



Technical Skills

- Program Management tools (e.g., Jira)
- Dashboards (e.g., Power BI)
- Documentation and reporting tools



AI Systems/Business Analyst



Position Summary

The AI Systems / Business Analyst bridges business objectives and AI delivery teams by translating strategic and operational needs into clear, testable system requirements across data, models, workflows, and governance. The role ensures AI solutions are well-scoped, feasible, and aligned to user needs and business outcomes, producing requirements, user journeys, and acceptance criteria that guide developers, data teams, and product owners. It delivers clarity, traceability, and quality across the AI lifecycle, partnering closely with business stakeholders, solution product owners, data and AI engineers, architects, and governance teams to enable scalable, responsible, and high-value AI solutions.



Job Duties and Essential Functions

- Translates business needs into structured AI system requirements covering data, models, workflows, and governance.
- Creates user stories, process maps, acceptance criteria, and test scenarios aligned to solution outcomes.
- Partners with product owners, engineers, and data teams to validate feasibility, dependencies, and solution design.
- Ensures completeness, accuracy, and Responsible AI compliance across requirements and testing artifacts.
- Supports test planning, user validation, and quality checks across AI features, agents, and workflows.
- Facilitates stakeholder alignment, readiness, and change enablement to drive successful solution adoption.



Functional Skills

- AI requirements analysis
- AI use-case scoping and problem framing
- Acceptance criteria and quality definition for AI systems
- Cross stakeholder alignment and communication
- Data and insight interpretation for use case decision support
- User journey and AI interaction mapping
- Workflow and process modeling for AI solutions
- Responsible AI



Technical Skills

- Requirements documentation tools (e.g., Confluence)
- Process mapping and system diagramming (e.g., Lucidchart)
- Journey mapping and experience visualization tools (e.g., Figma)
- Gap analysis and dependency mapping tools (e.g., spreadsheets)
- Basic data analysis and interpretation tools (e.g., BI)



AI Engineer



Position Summary

The AI Engineer designs, builds, and operates AI systems that enable scalable, secure, and reliable AI capabilities across the enterprise. This role focuses on developing and integrating AI-driven services, including agent-based workflows, model integrations, orchestration, and , to support both shared platforms and business solutions. The AI Engineer applies strong software engineering practices to ensure AI solutions are production-ready, governed, and aligned with enterprise standards. This role partners closely with product managers, architects, security, and operations teams to deliver AI capabilities that support business outcomes and responsible AI adoption.



Job Duties and Essential Functions

- Designs, develops, and maintains AI systems and services, including agent workflows, orchestration, and model integrations
- Implements prompt, context, memory, and tool management to enable consistent and reusable AI behavior
- Builds and supports telemetry, logging, and for AI workloads in production.
- Integrates AI capabilities with enterprise systems, data platforms, and APIs.
- Ensures AI solutions meet scalability, reliability, security, and governance requirements.
- Collaborates with product, architecture, and operations teams to deliver roadmap capabilities.
- Supports testing, debugging, and continuous improvement of AI services



Functional Skills

- Agentic AI system design and orchestration
- AI , evaluation, and quality
- Agent and model lifecycle management
- Context engineering
- Generative AI and LLM application engineering
- LLM Ops/AgentOps
- Prompt engineering
- RAG, vector search, and knowledge engineering
- Responsible AI, governance, and security



Technical Skills

- Generative AI development (e.g., Gemini Enterprise on GCP)
- Prompt, context, memory, and tool orchestration understanding
- Retrieval-Augmented Generation (RAG) and vector search tools (e.g., Gemini Enterprise)
- Cloud-native AI architecture (e.g., GCP)
- API-first enterprise integration tools (e.g., Apigee, Kong)
- AI and evaluation tools (e.g., Trulens)



AI Taxonomy & Ontology Librarian



Position Summary

The AI Taxonomy & Ontology Librarian defines and maintains controlled vocabularies, taxonomies, and ontologies to ensure consistent data labeling, semantic clarity, and effective reasoning across AI systems. The role shapes the semantic structures that underpin model understanding, retrieval, context, and governance, enabling AI solutions to operate with accuracy, consistency, and compliance. It delivers high-quality knowledge assets, metadata standards, and semantic models while partnering closely with data teams, AI engineers, product teams, and governance stakeholders to ensure alignment, interoperability, and responsible use across the enterprise.



Job Duties and Essential Functions

- Designs, curates, and maintains taxonomies, ontologies, vocabularies, and metadata standards for AI systems.
- Ensures semantic consistency across data, models, agents, retrieval systems, and governance workflows.
- Partners with data, AI engineering, and product teams to integrate ontologies into pipelines, models, and systems.
- Defines labeling schemas, annotation guidelines, and semantic governance practices.
- Validates and improves semantic structures based on user feedback, model performance, and business needs.
- Ensures alignment with Responsible AI, compliance, traceability, and enterprise data standards.



Functional Skills

- AI ontology development and governance
- AI knowledge management
- Content engineering
- Data labeling schema design
- Information architecture and knowledge modeling
- Knowledge engineering
- Memory and context structuring for AI systems
- Model context protocol
- Semantic modeling and metadata governance
- Taxonomy and ontology management



Technical Skills

- Semantic model and knowledge graph design tools (Neo4j)
- Context structuring for LLM and agent tools (LangChain, Google ADK)
- Ontology-driven retrieval and embedding tools (Vector Search, memo)
- Enterprise metadata modeling and catalog tools (Dataplex)



AI UX Designer



Position Summary

The AI UX Designer designs the end-to-end user experience for AI-enabled systems, with a specific focus on creating defensive UI patterns that guide safe, responsible, and intuitive interactions between humans and AI agents. The role shapes user journeys, interface behaviors, guardrail experiences, and multimodal interaction flows that ensure clarity, trust, and seamless task execution. It delivers user-centered designs, prototypes, and interaction standards while partnering closely with product owners, solution teams, engineers, data scientists, and governance stakeholders to ensure AI solutions are usable, compliant, and aligned with user needs and enterprise expectations.



Job Duties and Essential Functions

- Designs user journeys, wireframes, and interaction flows for AI-enabled interfaces, with emphasis on defensive UI patterns.
- Translates user needs, risks, and AI behaviors into intuitive, safe, and guided UX designs.
- Partners with product, engineering, and data teams to refine agent workflows, prompts, responses, and UI treatments.
- Ensures UI patterns support Responsible AI, error prevention, transparency, and safety-by-design principles.
- Creates prototypes, usability tests, and experience guidelines for AI-based interactions.
- Iterates designs based on user feedback, model behavior insights, and solution performance.



Functional Skills

- AI Experience and Interaction Design
- AI user journey and task flow design
- Defensive UX Patterns for Human-Agent Collaboration
- Error handling, uncertainty, and fallback UX design
- Multimodal and Conversational Interaction Design
- Responsible-AI and guardrail experience design
- User-centered design validation and iteration
- UX standards definition for AI



Technical Skills

- Cloud-native AI architecture knowledge (e.g., GCP)
- Design tool proficiency e.g., Figma
- Accessibility and inclusive design for AI interfaces (WCAG guidelines)
- UX instrumentation and insight-driven design refinement (Generic analytics standards)





Position Summary

The AI UX Developer builds and maintains the user-interface components and front-end services that implement the AI user experience, translating UX designs into performant, secure, and scalable interfaces. The role turns interaction patterns (including defensive UI for human-agent workflows) into reusable components and application flows, integrates with AI platform capabilities and APIs, and ensures quality, accessibility, and responsible-AI guardrails in the UI. It delivers production-ready front-end features, design-system contributions, and telemetry-backed improvements, partnering closely with UX designers, product managers, AI/solution engineers, platform teams, data and security stakeholders to drive adoption and measurable outcomes.



Job Duties and Essential Functions

- Implements user journeys and defensive UI patterns as reusable, accessible front-end components.
- Integrates UI with AI services (prompts, tools, agents, retrieval) via secure APIs and client gateways.
- Ensures front-end performance, reliability, and with quality and accessibility baked in.
- Collaborates with UX, product, and engineering to refine requirements, states, and error/edge behaviors.
- Applies Responsible-AI controls in UI (disclosures, rate/usage feedback, content handling and guardrails).
- Contributes to design systems, documentation, and continuous improvement through usage analytics.



Functional Skills

- AI experience design and development
- Human-Agent interaction implementation
- Defensive UI pattern Implementation for AI systems
- Frontend quality standards definition
- Responsible-AI guardrail enablement in UI
- Telemetry-driven UX optimization
- Design system contribution and reuse
- Cross-functional UX delivery enablement



Technical Skills

- Cloud-native AI architecture knowledge (e.g., GCP)
- Proficiency with CSS, JavaScript, Pattern Libraries, Version Control Systems



AI Solution Architect



Position Summary

The AI Solution Architect designs end-to-end AI solutions for specific products and use cases, ensuring they are scalable, secure, and aligned with enterprise architecture, platform capabilities, and governance standards. The role translates business and functional needs into solution architectures spanning data flows, model integration, agent behaviors, retrieval patterns, UI interactions, and operational requirements. It delivers solution blueprints, integration patterns, and technical guidance that enable predictable, compliant implementation while partnering closely with product managers, AI engineers, data teams, UX, cloud, security, and platform owners to ensure seamless delivery and long-term maintainability.



Job Duties and Essential Functions

- Designs end-to-end AI solution architectures across data, models, agents, workflow, and UI layers.
- Ensures solutions align with enterprise architecture, platform standards, security, and Responsible AI requirements.
- Defines integration patterns, component interactions, data contracts, and runtime flows.
- Partners with engineering, data, UX, and platform teams to validate feasibility and break down solution requirements.
- Oversees technical quality across design, build, performance, compliance, and operational readiness.
- Provides architectural guidance throughout delivery, resolving risks, dependencies, and design decisions.



Functional Skills

- End-to-End AI solution architecture design
- Agent Ops/LLM Ops
- Agent, model & retrieval architecture patterns
- Business-to-AI requirement translation
- Enterprise integration & API design patterns
- Multi-agent orchestration architecture
- Security, governance & compliance alignment
- Scalability, performance & cost architecture
- Operational readiness & lifecycle architecture



Technical Skills

- Agentic AI and LLM System Architectures (e.g., Gemini Enterprise, Google ADK)
- Model Integration and Serving Architecture (e.g., Vertex AI)
- Retrieval-Augmented Generation (RAG) Design tools (e.g., Vector Search, Dataplex)
- Data Flow, Context, and Knowledge tools (e.g., Dataplex, Neo4J)
- Enterprise API and Integration tools (e.g., APIGEE, Kong)
- Operational tools (e.g., Arize, Grafana)



AI Quality Engineer



Position Summary

The AI Quality Engineer ensures AI platform components and AI-enabled solutions meet the highest standards of quality, safety, and reliability through both automated and manual testing—including adversarial testing and red-teaming. The role validates functionality, performance, data integrity, and AI/agent behavior across models, retrieval systems, and orchestration flows. It delivers test frameworks, evaluation harnesses, guardrail validations, and telemetry-driven insights, partnering closely with product owners, engineers, data teams, architects, and governance stakeholders to ensure AI solutions are robust, compliant, and ready for production at scale.



Job Duties and Essential Functions

- Designs and executes functional, behavioral, red-team, and adversarial tests across AI models, agents, and pipelines.
- Builds automated test harnesses for prompts, tools, reasoning steps, retrieval flows, and integrations.
- Validates performance, latency, reliability, and model/tool selection behavior under different workloads.
- Ensures data integrity, safety, compliance, and Responsible AI guardrails are consistently upheld.
- Partners with engineers, data teams, and product owners to identify defects, assess risks, and drive quality improvements.
- Monitors telemetry and evaluation metrics to guide continuous testing, drift detection, and stability improvements.



Functional Skills

- AI evaluation, red-team & adversarial testing patterns
- Agent and model behavior validation
- Agent monitoring
- AI production quality gates and acceptance criteria
- AI quality assurance and release
- Guardrail, safety & policy compliance validation
- Risk-based quality assessment for AI systems
- Retrieval, context & data integrity testing for AI
- Telemetry-driven quality analysis



Technical Skills

- Automated and manual testing systems
- AI evaluation, scoring, and regression testing tools (e.g., TruLens, RAGAS, DeepEvals)
- AI platform reliability (e.g., GCP Storage and Compute Stack)



AI DevOps Engineer



Position Summary

The AI DevOps Engineer builds and operates the CI/CD, infrastructure, and automation that deploy, monitor, and scale AI systems reliably and securely. The role translates platform and solution requirements into resilient pipelines, runtime environments, , and safety controls across models, agents, retrieval, and data flows. It delivers production-ready environments, golden-path release processes, and telemetry-driven operations, partnering closely with AI engineers, solution architects, platform teams, security, and SRE/operations to ensure predictable, compliant, and cost-efficient delivery at scale.



Job Duties and Essential Functions

- Designs and maintains CI/CD pipelines for models, agents, services, and prompts; automates testing, security, and policy gates.
- Provisions and operates runtime infrastructure (APIs, gateways, storage, vector/search,) with reliability and scale.
- Implements runtime telemetry, tracing, metrics, and alerting for AI pipelines; drives SLOs, error budgets, and incident response.
- Enforces security, privacy, and Responsible AI controls across build, deploy, and run (secrets, access, content safety, audit).
- Optimizes performance and cost through caching, routing, autoscaling, and environment right-sizing.
- Collaborates with product, architecture, data, and platform teams to standardize golden paths and reusable delivery patterns.



Functional Skills

- AI CI/CD pipeline operations and release management
- Agent and model lifecycle operations
- Automation and deployment of AI systems
- Cost and performance management for AI platforms and solutions
- LLMOPs and Agent Ops
- Operational coordination across AI, platform, and SRE teams
- Operational governance and compliance Production environment provisioning for AI workloads
- Runtime and telemetry integration for AI systems



Technical Skills

- AI CI/CD pipeline implementation and automation (e.g., Vertex AI)
- Agent and model deployment orchestration tools (e.g., Vertex AI, Google ADK)
- Infrastructure provisioning and environment management (e.g., GCP Storage and Compute Stack)
- and telemetry integration tools (e.g., Grafana, Dynatrace)



AI Platform Operations Engineer



Position Summary

The AI Platform Operations Engineer supports, maintains, and operates the enterprise AI platform to ensure reliable, secure, and efficient AI capabilities across the organization. This role focuses on day-to-day platform operations, including system monitoring, incident resolution, patching, version upgrades, and support of AI services such as models, agents, orchestration layers, retrieval systems, and APIs. The AI Platform Operations Engineer applies strong full-stack and platform engineering practices to keep AI environments stable, performant, and compliant, partnering closely with AI engineers, platform architects, SRE, security, and product teams to resolve issues, reduce operational risk, and ensure the AI platform remains production-ready and aligned with enterprise standards



Job Duties and Essential Functions

- Supports and maintains AI platform services, including models, agents, orchestration, retrieval, memory, and APIs.
- Performs platform patching, version upgrades, configuration changes, and dependency management to ensure security and stability.
- Monitors platform health and performance; investigates, troubleshoots, and resolves incidents, alerts, and support tickets.
- Diagnoses and resolves issues across application, API, data, and infrastructure layers of the AI platform.
- Implements operational fixes, automation, and runbooks to improve reliability and reduce recurring issues.
- Collaborates with AI engineers, SRE, security, and platform teams to support releases, changes, and continuous platform improvement.



Functional Skills

- AI platform operations
- AI platform service management
- Cross-functional engineering collaboration
- Full-stack troubleshooting
- Incident response
- Issue triage and root cause analysis
- Operational diagnostics
- Operational governance and compliance controls
- Production environment provisioning for AI workloads
- Patch management
- Responsible AI



Technical Skills

- AI platform operations and runtime support tools (e.g., Vertex AI, Gemini Enterprise)
- Full-stack applications (e.g., Python, APIs, services)
- Cloud-native platform (e.g., GCP Storage and Compute Stack)
- API management and integration support tools (e.g., APIGEE, Kong)
- Retrieval, memory, and context system (e.g., Vector Search, Redis Cache / Cloud Memorystore)
- Operational automation and scripting (e.g., Python)



AI Site Reliability Engineer (SRE)



Position Summary

The AI SRE ensures the reliability, scalability, and performance of AI platforms and solutions by monitoring, maintaining, and optimizing production systems across models, agents, retrieval, data flows, and APIs. The role builds operational guardrails, foundations, resiliency patterns, and automated recovery mechanisms that keep AI systems consistent and predictable in production. It delivers high-quality runtime environments, incident insights, and performance improvements while partnering closely with AI engineers, solution architects, platform teams, security, and product owners to increase reliability, reduce operational risk, and evolve system health over time.



Job Duties and Essential Functions

- Monitors AI systems end-to-end (agents, models, retrieval, APIs) and ensures availability, performance, and resilience.
- Implements SLOs, SLIs, alerting strategies, incident response plans, and automated remediation workflows.
- Optimizes system performance through tuning, caching, load distribution, and architectural improvements.
- Ensures operational compliance with security, privacy, governance, and Responsible AI controls.
- Partners with engineering and architecture teams to embed reliability patterns into design and development.
- Conducts root-cause analyses, drives continuous improvement, and contributes to operational playbooks and runbooks.



Functional Skills

- AI production reliability engineering
- Agent monitoring
- Automation development for risk mitigation
- Continuous service improvement and resilience optimization
- Cross-functional runtime collaboration and communication
- Incident response and root-cause analysis
- Reliability standards and SLO definition for AI systems
- Scalability and capacity planning for AI workloads



Technical Skills

- AI platform reliability (e.g., GCP Storage and Compute Stack)
- Agent and model monitoring and alerting systems (e.g., Arize, Dynatrace)
- End-to-end and tracing tools (e.g., Grafana)
- API and integration tools (e.g., APIGEE, Kong)



AI Security Architect



Position Summary

The AI Security Architect defines and enforces AI-specific security controls across data, models, agents, workflows, APIs, and infrastructure to ensure AI systems are secure, compliant, and resilient against misuse. The role establishes enterprise guardrails, governance patterns, and architectural controls that address data protection, access management, model security, content safety, threat mitigation, and regulatory compliance. It delivers security design patterns, risk assessments, approval frameworks, and integration standards, partnering closely with platform teams, solution architects, AI engineers, data/security teams, governance functions, and product stakeholders to embed security-by-design into all AI solutions.



Job Duties and Essential Functions

- Defines enterprise AI security architecture, controls, guardrails, and policies across data, models, agents, and workflows.
- Conducts threat modeling, risk assessments, and security reviews for AI platforms and solutions.
- Establishes secure integration, access, and model-usage patterns (including prompt, tool, and agent security).
- Implements monitoring, detection, and guardrail frameworks to ensure safe and compliant AI behavior.
- Partners with engineering, architecture, data, cloud, and governance teams to embed security-by-design.
- Oversees compliance with privacy, regulatory, Responsible AI, and audit-readiness requirements.



Functional Skills

- AI security, governance, risk assessments
- Agent governance – registry certification
- Agent security
- Cross-functional security leadership and stakeholder alignment
- Regulatory, privacy, and compliance alignment for AI systems
- Security-by-design for AI
- Threat modeling and misuse risk management for AI and agents



Technical Skills

- AI and agent security architecture tools (e.g., Google ADK, Vertex AI)
- Data protection and leakage prevention tools (e.g., Dataplex, Model Armor)
- Identity and access management tools (e.g., APIGEE, Kong)
- Model, prompt, and inference security tools (e.g., Model Armor)
- Secure APIs and workflow orchestration tools (e.g., APIGEE)
- AI threat detection and incident response tools (e.g., Arize, Dynatrace)
- Runtime guardrails and content safety tools (e.g., Trulens, RAGAS/DeepEvals)



Model Risk Manager



Position Summary

The Model Risk Manager oversees the identification, assessment, and mitigation of risks associated with AI and ML models across their lifecycle, ensuring safe, reliable, and compliant use of AI throughout the enterprise. The role defines risk-control frameworks, validates model and agent behaviors, monitors drift and performance, and enforces governance expectations spanning data quality, explainability, robustness, fairness, and regulatory compliance. It delivers structured risk assessments, model validation documentation, and mitigation plans while partnering closely with AI engineers, data teams, architects, governance leads, security, legal/compliance, and product stakeholders to ensure AI systems operate within acceptable risk thresholds.



Job Duties and Essential Functions

- Conducts end-to-end risk assessments for AI/ML models and agents, covering data, logic, behavior, and operational risk.
- Establishes model validation frameworks, tests, controls, and approval mechanisms aligned with governance and policy.
- Evaluates model robustness, fairness, safety, reliability, explainability, and compliance with regulatory requirements.
- Monitors model performance, drift, telemetry, and incidents; drives remediation and continuous improvement.
- Partners with engineering, data, security, governance, and product teams to embed risk controls across the lifecycle.
- Creates and maintains risk documentation, audit evidence, mitigation plans, and model governance artifacts.



Functional Skills

- AI and model risk framework definition and ownership
- AI lifecycle risk assessment and control enforcement
- AI risk issue remediation and mitigation planning
- AI regulatory, compliance, and audit readiness management
- Agent governance
- Clear risk communication to technical and non-technical leaders
- Cross-functional risk governance and stakeholder coordination
- Model validation standards and approval processes
- Responsible AI



Technical Skills

- Model risk assessment and validation tools (e.g., RAGAS/DeepEvals)
- Agent behavior testing and control validation tools (e.g., Trulens)
- Model performance, drift, and stability monitoring tools (e.g., Arize)



Observability Engineer



Position Summary

The Observability Engineer designs and operates end-to-end telemetry, tracing, and evaluation systems that monitor model and agent behavior—capturing prompts, context, tool calls, memory, performance, safety signals, and cost metrics—to ensure AI systems remain reliable, explainable, and efficient in production. The role builds diagnostic tooling, insight pipelines, and run-time instrumentation that strengthen debugging, optimization, and trust in AI. It partners closely with AI engineers, SRE, data teams, architects, product owners, and governance stakeholders to deliver clarity, stability, and operational resilience across AI workloads.



Job Duties and Essential Functions

- Designs, builds, and maintains telemetry and observability infrastructure for AI models, agents, and workflows.
- Captures and analyzes behavioral and performance signals including prompts, latency, errors, memory, and tool calls.
- Implements tracing, logging, dashboards, metrics, alerting, and continuous monitoring capabilities.
- Partners with AI engineering, SRE, and platform teams to diagnose issues, optimize performance, and improve reliability.
- Ensures observability systems meet Responsible AI, safety, compliance, and governance requirements.
- Delivers insights, evaluation reports, and recommendations for ongoing improvements.



Functional Skills

- AI Behavior Analysis & Root-Cause Diagnostics
- AI Observability Strategy & Signal Design
- Governance & Safety Signal Integration
- Performance Characterization & Bottleneck Detection
- Reliability Improvement & Incident Insight Generation
- Traceability, Explainability & Transparency Enablement
- Defining Operational Health Metrics



Technical Skills

- APIGEE
- Arize
- Dataplex
- Distributed Tracing (OpenTelemetry, Jaeger, Zipkin)
- Dynatrace
- Grafana
- Kong
- LiteLLM
- Log and Metrics Pipeline Engineering (ELK, Cloud Logging, Prometheus)
- Model Armor
- Neo4j
- RAGAS / DeepEvals
- Real-Time Monitoring and Alerting System Design
- Redis Cloud Memory Store
- Streaming and Event Telemetry (Pub/Sub, Kafka)
- TruLens
- Vector Search
- Vertex AI and Azure OpenAI Telemetry Awareness

